

Unveiling Starlink for PNT

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Abstract

This paper provides a comprehensive theoretical and experimental description of how to exploit Starlink low Earth orbit (LEO) satellites for positioning, navigation, and timing (PNT). First, the paper reveals for the first time, the *full* Starlink orthogonal frequency division multiplexing (OFDM) beacon, which spans the whole time-frequency resource grid. This description of the beacon is achieved through *blind* beacon estimation, which shows that the Starlink sequences published in the literature only comprise 0.66% of Starlink's *full* OFDM. Exploiting this full OFDM beacon is shown to increase the receiver's process gain by nearly 18 dB compared to only using signals published in the literature. This process gain, in turn, unlocks higher effective SNR at the receiver's correlator output, enabling reliable acquisition and tracking in low SNR regimes imposed by using low-gain antennas. Second, the paper studies and compares the maximum achievable received carrier-to-noise density ratio (C/N_0) for different reception scenarios. Third, the paper shows the first experimental results of navigation observables extracted using OFDM signals transmitted by Starlink satellites, namely the carrier phase, Doppler shift, and code phase. The paper provides the most comprehensive Starlink signal collection from 2021 through 2024 and analyzes the quality of pilot-tone versus OFDM-based observables. Results show that step-like corrections sometimes contaminate all the OFDM-based navigation observables from Starlink satellites, rendering their raw integration a challenge for precise positioning. Fourth, the paper shows how corrections made to the OFDM carrier frequency offset (CFO) can be estimated *on-the-fly* with a good degree of fidelity within the tracking loop of the software-defined receiver. Unlike the CFO corrections, the estimation of code phase corrections is shown to be intractable, rendering pseudoranges from Starlink signals not suitable for reliable positioning. Moreover, the tracked OFDM carrier phase revealed excessive slips due to the employed communication scheme. Finally, the paper demonstrates the first positioning solution that uses OFDM-based Doppler shift exclusively. Four positioning frameworks are formulated and assessed: (i) pilot tone-based Doppler shift tracking that exhibits no sign of contamination from the OFDM-related corrections, (ii) OFDM-based Doppler shift with uncorrected CFOs, (iii) OFDM-based Doppler shift with corrected CFOs that are estimated *on-the-fly*, and (iv) OFDM-based Doppler shift with corrected CFOs that are estimated using the knowledge of an assumed cooperative base station. The unprecedented results from these analyses show that, with an average of only

three active Starlink satellites, a positioning solution with a 3D position estimation error of two meters can be achieved in only 20 seconds.

Keywords

LEO, navigation, OFDM, PNT, signals of opportunity, Starlink, uncooperative satellites

1 | INTRODUCTION

The birth of low Earth orbit (LEO) satellite megaconstellations has resulted in tremendous interest in exploring the use of their signals for positioning, navigation, and timing (PNT) (Janssen et al., 2023; Jardak & Jault, 2022; Kassas, Khairallah et al., 2024; Kassas, Kozhaya, Kanj, et al., 2023; Khalife & Kassas, 2023; Menzione & Paonni, 2023; Prol et al., 2024, 2022, 2023; Stock et al., 2024). To this end, numerous studies over the past few years have explored various challenges in opportunistic PNT with LEO, from addressing space vehicle orbit, clock, and propagation errors (Cassel et al., 2022; Jiang et al., 2023; Kassas, Hayek et al., 2024; Khairallah & Kassas, 2024; Morton et al., 2022; Saroufim et al., 2023; Wang et al., 2023; Wu et al., 2023; Ye et al., 2023; Zhao et al., 2023); to receiver and signal design (Bilardi, 2021; Egea-Roca et al., 2022; Ferre et al., 2022; Huang et al., 2022; Kassas, Kozhaya, Saroufim, et al., 2023; Kassas et al., 2021; Pinell et al., 2023; Tan et al., 2019; Q. Wei et al., 2020; Yang et al., 2023); to analyzing PNT performance (Farhangian et al., 2021; Guo et al., 2023; Hartnett, 2022; Jiang et al., 2022; Kozhaya et al., 2021; Kozhaya & Kassas, 2024a; More et al., 2022; Sabbagh & Kassas, 2023; Shi et al., 2023; Singh et al., 2022).

With over 6,000 satellites already in LEO, SpaceX's Starlink is currently the leading provider of LEO-based broadband communication. Indeed, SpaceX is reshaping the communication sector with its commercial Starlink and military Starshield LEO; however, as of today, little to no effort is being invested to expand the constellation's capabilities to offer PNT services. Shortly after the first batch of Starlink satellites was launched, Starlink-related studies began to populate the literature, ranging from studies of the different aspects of the constellation to studies of how the constellation could be opportunistically exploited in applications other than those it was originally intended for.

The first works of note exploited the constellation for positioning purposes by tracking the carrier phase and Doppler shift of one of Starlink's persistent signals (referred to as the "beacon" in this paper) (Jardak & Adam, 2023; Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021). The beacon that these papers considered was a trail of nine unmodulated, data-less, pilot tones at the center of each Ku-band user downlink channel. Later, other studies examined the structure of the uplink (Stock et al., 2023) and downlink (Grayver et al., 2024; Humphreys et al., 2023; Kanj et al., 2023; Kozhaya et al., 2023; Neinavaie & Kassas, 2024a, 2024b) signals transmitted by Starlink to reveal salient information about their transmission, modulation, and multiple access schemes. These papers determined that Starlink employs orthogonal frequency division multiplexing (OFDM) for data modulation and exposed many signal parameters, from the frame duration to some decoded sequences transmitted by Starlink for synchronization purposes.

While some researchers focused on exploiting Starlink's beacons (pilot tones and OFDM) and turning them into signals of opportunity for PNT purposes, other efforts demonstrated the use of Starlink satellites as illuminators of opportunity for passive radar applications (Blazquez-Garcia et al., 2024; Blázquez-García et al.,

2022, 2023; del Hoyo et al., 2023). Other recent work has evaluated the performance of the constellation in its primary application domain, communication, by studying the physical layer transmission rates, throughput, and protocols used at the user equipment terminals (Garcia et al., 2024; Garcia et al., 2023).

These previous studies have only scratched the surface of using Starlink LEO satellite signals for PNT. To the authors' knowledge, no comprehensive study to date has showcased the intricacies of exploiting Starlink's signals for PNT: from the full OFDM frame, to the achievable carrier-to-noise density ratio (C/N_0), to the feasibility of sustainable code, carrier, and Doppler tracking. This paper provides such a comprehensive study and answers the following questions: (1) Can the pilot tones still be tracked using low-gain antennas as shown in (Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021)? (2) How reliable is the OFDM beacon published in (Humphreys et al., 2023) in low SNR regimes? (3) Are Starlink's pseudorange observables, similar to those shown in (Grayver et al., 2024), useful for positioning? (4) Is there more to exploit in the Ku-band downlink OFDM signals? (5) Will the navigation observables generated using the OFDM beacon unlock more precise positioning compared to its pilot tone-based counterpart?

Although some of these questions seem to have obvious answers based on the available literature, this paper shares some surprising results concerning the use of highly optimized communication signals for PNT. Specifically, this paper:

- Reveals, for the first time, the *full* Starlink OFDM beacon, which spans the whole time-frequency resource grid. This is achieved through *blind* beacon estimation, which shows that the Starlink sequences published in the literature only comprise 0.66% of Starlink's *full* OFDM. Exploiting the full OFDM beacon is shown to increase the receiver's process gain by nearly 18 dB compared to only using the signals published in (Humphreys et al., 2023). This process gain, in turn, unlocks a higher effective signal-to-noise ratio (SNR) at the receiver's correlator output, enabling practical exploitation of received signals for PNT in low SNR regimes with a commercial off-the-shelf low-noise block downconverter with feed horn (LNBF).
- Provides a comprehensive theoretical and experimental description for how to exploit Starlink satellites for PNT. The maximum achievable received C/N_0 is studied and compared for different scenarios, namely: (i) pilot tones versus OFDM-based beacons and (ii) low-gain versus high-gain reception setups.
- Shows the first experimental results of extracting navigation observables using the *OFDM signal* transmitted by Starlink satellites, namely the carrier phase, Doppler shift, and code phase. Moreover, the quality of the OFDM-based observables is discussed extensively and compared against the pilot tone-based observables. Aligned with the primary application of the constellation, which is providing broadband communication, the results show that, before the year 2024, step-like corrections contaminate all the OFDM-based navigation observables, rendering their raw integration a challenge for precise positioning. After 2024, fewer OFDM-related corrections are being recorded, promoting the constellation's ability to provide positions with meter-level accuracy.
- Shows how the consistent corrections made to the OFDM carrier frequency offset (CFO) can be estimated on-the-fly, with a good degree of fidelity, within the tracking loop of the software-defined receiver.
- Showcases the first full Ku-band downlink channel capture and examines the number of simultaneously active Starlink satellites. The paper shows that,

with an average of only three active Starlink satellites, a positioning solution with meter-level 3D position error can be achieved in just 10 seconds.

This paper is organized as follows. First, Section 2 gives an overview of the Starlink constellation. Section 3 then discusses the model of the received signal from Starlink satellites. Section 4 reveals the *full* Starlink OFDM beacon obtained through blind beacon estimation and discusses its time- and frequency-domain characteristics. Section 5 discusses the different signal reception setups and their expected received C/N_0 . Section 6 discusses the receiver design used to acquire and track Starlink satellites. Section 7 shows experimental results of all the different navigation observables that can be extracted from Starlink satellites and assesses their quality. Section 8 shows experimental results of different methods for positioning with Starlink satellites and assesses their accuracy. Finally, Section 9 gives concluding remarks.

2 | STARLINK CONSTELLATION OVERVIEW

There are currently more than 6,000 Starlink LEO satellites orbiting the Earth to provide broadband Internet coverage. The Starlink constellation resides at an altitude between 540 and 570 km and comprises different satellite shells with non-polar and near-polar orbits. Figures 1 and 2 show a heat map of the number of visible Starlink LEO satellites above a 10-degree elevation mask at any given time as of March, 1, 2024 (Celestrak, 2024) and after the deployment of the full, 30,000-satellite constellation (Space Exploration Holdings, 2021), respectively. An elevation mask of 10 degrees was chosen to filter out satellites with potentially distorted signals due to atmospheric effects, blockage from trees and buildings, and multipath. Figures 1 and 2 imply that, as of March 1, 2024, more than 50 satellites are present above any non-polar region at all times, and that this number is expected to increase to more than 100 after the full constellation is deployed. While Figures 1 and 2 show the total number of Starlink satellites above a given region at any time, only a subset of these satellites are simultaneously active to provide data for online subscribers. For example, Figure 3 presents an experimental data set showing the number of visible Starlink satellites for a receiver located in Columbus, OH, USA in July 2024. As shown in Figure 3, there is always an active Starlink satellite overhead, and, on average, three satellites are simultaneously active. Note that these numbers can depend on the surrounding user activity, location, and time at which the receiver is listening to these satellites, making the opportunistic Starlink PNT somewhat unpredictable. Nevertheless, the average number of active Starlink satellites is expected to rise as the number of subscribers increases and as the constellation is fully deployed.

The Starlink constellation uses the Ku-band to transmit its circularly polarized downlink signals to the user equipment. The Ku-band, which spans 10.7 to 12.7 GHz, is divided into eight 250 MHz channels. Moreover, the Starlink constellation realizes a hexagonal cell deployment with 48 simultaneous steerable spot-beams with a radius of at least 10 km (Blázquez-García et al., 2022). While the spotbeam of Starlink satellites is narrow, multiple satellites can simultaneously illuminate a given cell. This is achieved by employing different multiple-access techniques in the (i) time-domain, in which a subset of satellites transmit while the others are silent, (ii) frequency-domain, in which each satellite transmits at one of the eight different Ku-band downlink channels, (iii) polarization, in which the satellite uses the right- or left-hand circular polarization, and (iv) power-domain,

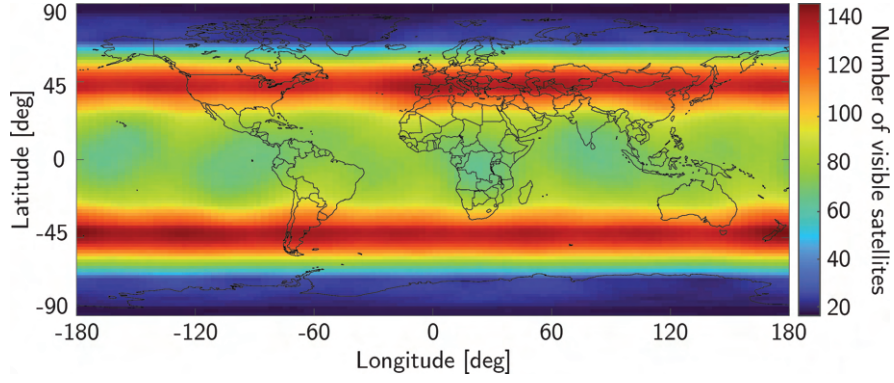


FIGURE 1 Heat map showing the number of visible Starlink LEO satellites above a 10-degree elevation mask at any point in time as of March 1, 2024.

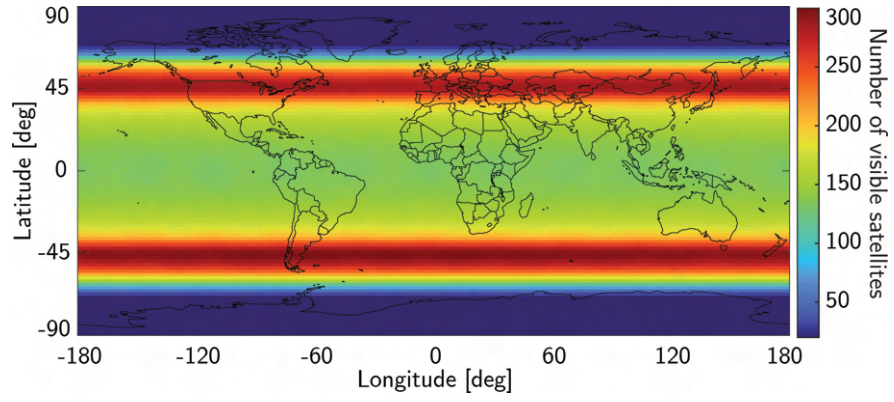


FIGURE 2 Heat map showing the predicted number of visible Starlink LEO satellites above a 10-degree elevation mask for the full, 30,000-satellite constellation at any point in time.

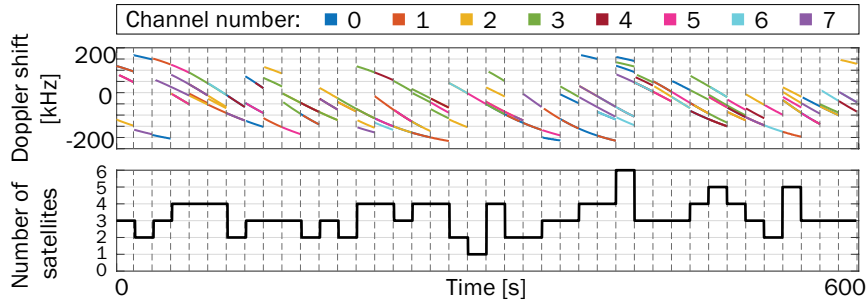


FIGURE 3 Experimental data set showing the number of visible Starlink satellites for a receiver located in Columbus, OH, USA in July 2024.

in which different power levels are allocated for different users. The parameters of the Starlink constellation are summarized in Table 1.

3 | SIGNAL MODEL

This section discusses the model of Starlink's Ku-band downlink signals. Consider $x(t) = \sqrt{C}s_{\text{III}}(t) + n_d(t)$ to be the baseband signal transmitted by a Starlink satellite, where C is the transmitter's carrier power in Watts, $s_{\text{III}}(t) \triangleq \sum_{k=-\infty}^{\infty} s(t - kT_0)$ is

TABLE 1
Parameters of the Starlink Constellation

Parameter	Value
Altitude [km]	540–570
Current number of satellites	6,219
Planned number of satellites	30,000
Downlink frequency [GHz]	10.7–12.7
Number of channels	8
Useful channel bandwidth [MHz]	240
Number of steerable spotbeams	48
Spotbeam radius [km]	≥ 10
Modulation	OFDM
Frame duration [ms]	4/3

the repetitive beacon stream, $s(t)$ is a deterministic beacon of duration T_0 seconds, and $n_d(t)$ is the random user data. Assume $x(t)$ being transmitted at a carrier frequency f_c that corresponds to one of the eight channels in the Ku-band. The transmitted signal is given by $x_c(t) \triangleq x(t) \exp[j2\pi f_c t]$. After propagating in a noisy channel, the signal reaches the receiver's antenna with an apparent delay in time and phase, denoted by $\tau(t)$ and $\delta_\theta(t)$, respectively. The received signal is given by $r_c(t) = x_c(t - \tau(t)) \exp[j\delta_\theta(t)] + n_c(t)$, where $n_c(t)$ is the channel noise modeled as a complex, zero-mean, white noise with two-sided power spectral density N_0 . After mixing with a local oscillator at a frequency $\hat{f}_c$, the received signal can be expressed as:

$$r(t) \triangleq r_c(t) \exp[-j2\pi \hat{f}_c t] = \sqrt{C} s_{\text{III}}(t - \tau(t)) \exp[j\theta(t)] + n(t),$$

where $\tau(t)$ is referred to as the code phase, $\theta(t) \triangleq \phi(t) + 2\pi \tilde{f}_c t$ is the beat carrier phase, $\phi(t) \triangleq -2\pi f_c \tau(t) + \delta_\theta(t)$ is the carrier phase, $\tilde{f}_c \triangleq f_c - \hat{f}_c$ is the CFO, and $n(t) \triangleq n_d(t - \tau(t)) \exp[j\theta(t)] + n_c(t) \exp[-j2\pi \hat{f}_c t]$ is the lumped user data and channel noise.

The code phase $\tau(t)$, in seconds, is modeled as:

$$\tau(t) \triangleq \frac{\rho(t)}{c} + \delta t_{\text{clk}}(t) + \delta t_{\text{atm}}(t) + v_\tau(t),$$

where $\rho(t)$ is the line-of-sight distance between the transmitter and the receiver, c is the speed of light, $\delta t_{\text{clk}}(t)$ captures the combined effect of the transmitter's and receiver's clock biases, $\delta t_{\text{atm}}(t)$ captures the atmospheric delay due to the ionosphere and troposphere, and $v_\tau(t)$ captures the lumped effect of multipath, channel noise, and other unmodeled errors.

At time $t_k \triangleq t_0 + kT_0$, where $k \in \mathbb{N}$ is the accumulation index and t_0 is some initial time defined to be 0, the k -th accumulation can be expressed as $r_k(t) w_{T_0}(t)$. Here, $r_k(t) \triangleq r(t + t_k)$ is a time-shifted version of the received stream and $w_{T_0}(t)$ is a window function that is unity within the interval $[0, T_0]$ and zero elsewhere.

Using the Taylor series expansion at time instant t_k , the carrier and code phases of the received signal can be expressed as:

$$\theta_k(t) \triangleq \theta(t + t_k) = \theta(t_k) + \dot{\theta}(t_k)t + \frac{1}{2}\ddot{\theta}(t_k)t^2 + \text{H.O.T.}, \quad (1)$$

$$\tau_k(t) \triangleq \tau(t + t_k) = \tau(t_k) + \text{H.O.T.} \approx \tau_k, \quad (2)$$

where $H.O.T.$ denotes the higher-order terms. By definition, $f_D(t) \triangleq \dot{\theta}(t)/2\pi$ is the apparent Doppler shift and $\dot{f}_D(t) \triangleq \ddot{\theta}(t)/2\pi$ is the apparent Doppler rate. The first-order term in the code phase is known as the Doppler stretch factor and manifests by stretching and dilating the received signal in the time-domain. By neglecting the Doppler stretch effect, the code phase can be approximated by its zeroth-order term for a coherent processing interval (CPI) of $T_0 = 4/3$ ms and a receiver bandwidth $B < 30$ MHz without inducing considerable power loss (less than a tenth of a dB) in the correlator output (Kozhaya & Kassas, 2024b). Note that while the full 240 MHz Starlink OFDM beacon, revealed in Section 4, was obtained by accounting for the Doppler stretch effect, the receiver design formulation and experimental results use a downsampled version of the 240 MHz beacon at 2.5 MHz and 5 MHz. Because these frequencies are less than 30 MHz, the Doppler stretch effect can be ignored in this case.

Using Equations (1) and (2), the signal at the k -th accumulation can be expressed as:

$$r_k(t) = \sqrt{C} s_{\text{III}}(t - \tau_k) \exp[j\theta_k(t)] + n_k(t),$$

where $n_k(t) \triangleq n(t + t_k)$ represents the lumped user data and channel noise.

For a given estimate of the code phase $\hat{\tau}_k$ and beat carrier phase $\hat{\theta}_k(t)$, the output at the k -th accumulation after modulating the incoming signal with a numerically-controlled oscillator is given by:

$$\tilde{r}_k(t) = r_k(t + \hat{\tau}_k) \exp[-j\hat{\theta}_k(t)] w_{T_0}(t) = \sqrt{C} s_{\text{III}}(t - \tilde{\tau}_k) \exp[j\tilde{\theta}_k(t)] w_{T_0}(t) + \tilde{n}_k(t), \quad (3)$$

where $\tilde{\tau}_k \triangleq \tau_k - \hat{\tau}_k$ is the code phase error, $\tilde{\theta}_k(t) \triangleq \theta_k(t) - \hat{\theta}_k(t)$ is the beat carrier phase error, and $\tilde{n}_k(t) \triangleq n(t + t_k + \hat{\tau}_k) \exp[-j\hat{\theta}_k(t)] w_{T_0}(t)$.

The estimates of the code and beat carrier phase that are fed to the numerically-controlled oscillator are obtained from: (i) a determined set of search values when the receiver is in the acquisition stage or (ii) the tracking loops when the receiver is locked to the source. The operations in Equation (3) are referred to as the code and carrier wipe-off.

After sampling at a rate of $F_s = 1/T_s$, the received signal at the k -th accumulation is given by:

$$r_k[n] = (s \otimes \tau_k)[n] \exp[j\theta_k[n]] + n_k[n], \quad n \in \mathbb{N},$$

where $(y \otimes \tau)[n] \triangleq \sum_{k=-\infty}^{\infty} y(t - kT_0 - \tau) w_{T_0}(nT_s)$ denotes the discrete-time circular shift operation of $y(t)$ by τ seconds, $\theta_k[n] \triangleq \theta_k(nT_s)$ is the discrete-time carrier phase sequence, and $n_k[n] \triangleq n_k(nT_s)$ is the discrete-time noise sequence. Note that $s[n] \triangleq (s \otimes 0)[n]$ is the discrete-time version of the beacon with length of $\frac{T_0}{T_s}$ samples. Finally, the discrete-time version of $\tilde{r}_k(t)$ is given by:

$$\tilde{r}_k[n] = (s \otimes \tilde{\tau}_k)[n] \exp[j\tilde{\theta}_k[n]] + \tilde{n}_k[n], \quad (4)$$

where $\tilde{\theta}_k[n]$ is the discrete-time carrier phase error sequence and $\tilde{\tau}_k$ is the code phase error.

4 | THE STARLINK OFDM BEACON

This section reveals, for the first time, the *full* Starlink OFDM beacon obtained through blind beacon estimation. The time- and frequency-domain characteristics of this beacon are also discussed here.

4.1 | Blind Beacon Estimation

Turning virtually any radio frequency (RF) source into a potential signal of opportunity for PNT requires knowledge of some persistent feature(s) in the source's transmitted signals. These features are referred to as the beacon. Current and future communication providers, whether they use terrestrial or non-terrestrial networks, are welcoming the adoption of reconfigurable software-defined transceivers that can readily change the structure of their transmitted signals (Lin et al., 2021; López et al., 2022; Z. Wei et al., 2021). However, some of these private networks, particularly Starlink, are not required to publicly share the structure of their proprietary transmitted signals. The fact that these networks use an undisclosed signal structure and can alter this structure without prior notice necessitate the development of a PNT framework that can blindly estimate signal-of-opportunity beacons.

The blind beacon estimator, similar to (Kozhaya et al., 2023), works by cycling between two steps until it achieves convergence. The first step is beacon detection, in which the estimator tests for the presence of a beacon based on an estimated C/N_0 . If a beacon is detected, the second step of the framework projects the current noisy observation of the beacon onto its best estimate of the beacon to estimate the time-varying parameters modulating the beacon: namely the carrier phase, frequency, and code phase. These time-varying parameters are then wiped from the current noisy observation of the beacon. Finally, the estimator uses the corrected observation to update the beacon estimate and then repeats this process until the energy of the beacon converges.

While the blind beacon estimator (Kozhaya et al., 2023) is outside the scope of this paper, this subsection demonstrates its efficacy in estimating the *full* OFDM beacon of Starlink's Ku-band downlink signal. In previous work, (Humphreys et al., 2023) only revealed the first two symbols of the OFDM beacon, the primary synchronization signal (PSS) and secondary synchronization signal (SSS), using a semi-blind, reverse engineering approach. Likewise, (Neinavaie & Kassas, 2024b) only revealed the existence of OFDM subcarriers and the beacon length. Here, this paper shows the *full* OFDM beacon.

To this end, an LNBF (RemoteQTH, 2024) connected to a 60 cm 35 dBi parabolic dish was used to listen to overhead Starlink satellites. The signal was recorded for one second with a Universal Software Radio Peripheral (USRP) X410 (Ettus Research, 2024) with a receiver bandwidth of 500 MHz. The USRP X410 was connected to a Global Positioning System (GPS) antenna to discipline its clock, which in turn, disciplines the LNBF oscillator. The recorded signal was stored for offline processing, in which the blind beacon estimator was applied to the captured signal to estimate the full 240 MHz OFDM beacon.

The block diagram of the high-gain capture system is shown in Figure 4.

The frame of the final estimated OFDM beacon is shown in Figure 5. This figure shows that the repetitive sequences (represented in yellow) sparsely span the whole time-frequency grid of the OFDM frame. Moreover, Figure 5 indicates the existence of two specific regions where the repetitive sequences are dense:

- **B1:** The beginning and end of the frame, spanning the whole 240 MHz channel bandwidth but not the whole beacon duration $T_0 = 4/3$ ms.
- **B2:** The low-side and high-side bands of the channel, spanning a 2 MHz bandwidth and the whole beacon duration.

The projection of the OFDM into the time domain shows how the time-domain power distribution of the beacon spans the whole beacon duration. Moreover, the frequency projection shows how both ends of the channel spectrum contain a powerful signal, which is exploited in this paper for PNT applications. Finally, the bottom plot in 5 shows a zoomed-in view of some OFDM symbols at the beginning and end of the frame. While the percentage of symbols exploited in (Grayver et al., 2024; Humphreys et al., 2023) is only $2/302 = 0.66\%$ (PSS and SSS), the full OFDM beacon shown in Figure 5 exploits 100% of the symbols.

After reconstructing the OFDM frame from the time-domain waveform of the beacon, the first three symbols were demodulated. The in-phase and quadrature (IQ) plots for these three symbols are shown in Figure 6(a)–(c). The first two blindly estimated symbols were compared with their counterparts published in

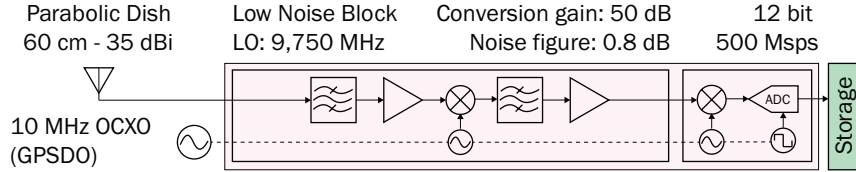


FIGURE 4 Hardware setup used for Starlink's high-gain signal capture and blind beacon estimation.

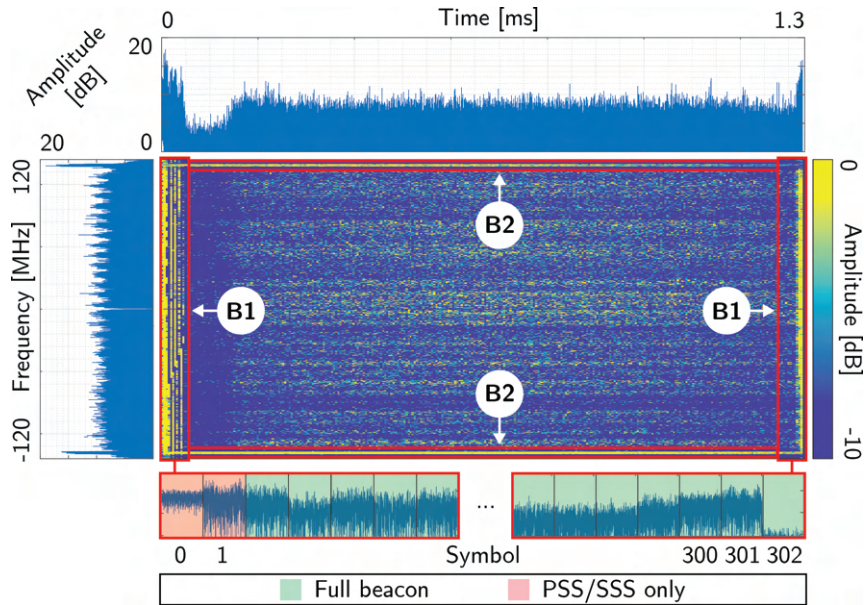


FIGURE 5 *Middle:* Time-frequency plot of Starlink's OFDM beacon. *Top:* Projection of the OFDM frame into the time domain. *Left:* Projection of the OFDM into the frequency domain. *Bottom:* Zoomed-in view of the time-domain waveform of some OFDM symbols at the beginning and end of the frame. The full OFDM beacon spans 100% of the symbols. The repetitive sequences are dense in two specific regions: (B1) the beginning and end of the frame, spanning the whole 240 MHz channel bandwidth but not the whole beacon duration $T_0 = 4/3$ ms, and (B2) the low-side and high-side bands of the channel, spanning a 2 MHz bandwidth and the whole beacon duration.

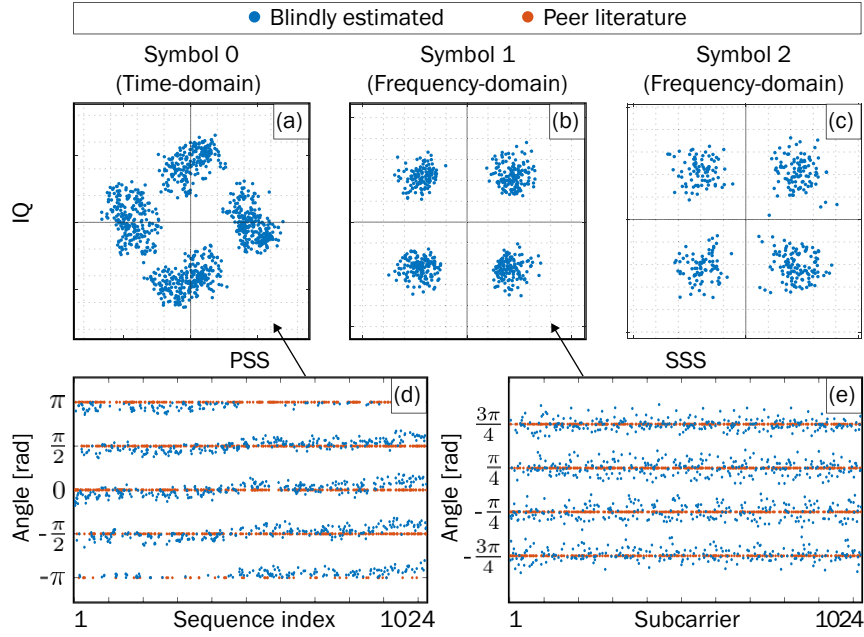


FIGURE 6 (a)–(c) In-phase and quadrature plot of the first three symbols after demodulation. (d) and (e) Comparison between the blindly estimated PSS and SSS sequences (in blue) and their exact values reconstructed from peer-reviewed literature (Humphreys et al., 2023) (in orange). The comparison had a 0% demodulation error.

(Humphreys et al., 2023) as the PSS/SSS, and the results showed an excellent match. Figure 6(d)–(e) shows this comparison between the blindly estimated PSS and SSS sequences (in blue) and their exact values reconstructed from peer-reviewed literature (in orange).

Although a high-gain setup was used for blind beacon estimation of Starlink’s OFDM beacon, the data presented in the rest of this paper relies solely on low-gain LNBFs. Here, the use of any bandpass-filtered version of the frame shown in Figure 5 is referred to as the *full* beacon because it spans nearly the entire beacon duration T_0 (i.e., 302 active symbols).

4.2 | Characteristics of the Starlink OFDM Beacon

Exploiting the full Starlink OFDM beacon has multiple advantages:

- More bandwidth implies improved resolution in the delay-domain, which manifests in a narrower main lobe in the auto-correlation function (ACF) of the beacon.
- More symbols being exploited (i.e., 100%) implies a longer integration time in the interval $[0, T_0]$, leading to improved resolution in the Doppler domain and better carrier tracking performance.
- Finally, the exploitation of additional resources, whether in the time or frequency domains, increases the processing gain of the receiver and enables it to perform under low SNR regimes, as will be discussed in Section 5.4.

Figure 7 shows the ACF of the blindly estimated Starlink OFDM beacon for receiver bandwidth ranging from 2.4 MHz to the full 240 MHz. At 240 MHz, the first expected null in the ACF is at $\frac{1}{240 \text{ MHz}} \approx 4 \text{ ns}$, implying that receivers exploiting the whole 240 MHz bandwidth and sensitive delay-locked loops are expected

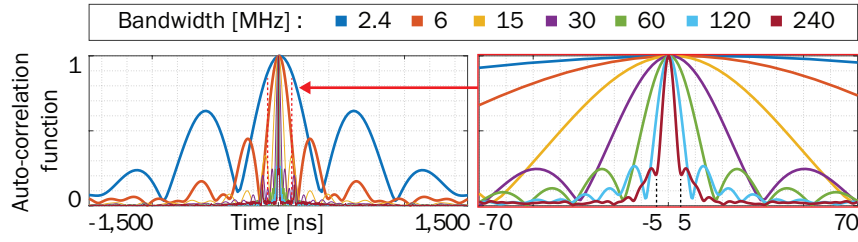


FIGURE 7 Auto-correlation function of the blindly estimated Starlink OFDM beacon versus receiver bandwidth.

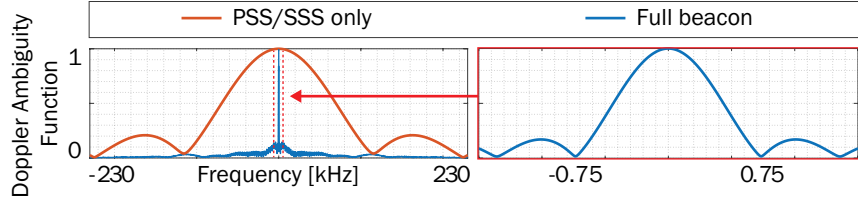


FIGURE 8 *Left:* Doppler ambiguity function of the OFDM beacon in (Humphreys et al., 2023), consisting of PSS/SSS only. *Right:* Doppler ambiguity function of the full Starlink OFDM beacon.

to achieve sub-nanosecond signal synchronization accuracy. The results shown in this paper are based on receiver bandwidths of 2.5 MHz and 5 MHz, leading to synchronization accuracy of 400 and 200 ns, respectively.

In the Doppler domain, having the full OFDM beacon spanning the whole T_0 results in an expected first null at $\frac{1}{1.33 \text{ ms}} \approx 750 \text{ Hz}$, which is shown in the right plot of Figure 8. Compared to (Humphreys et al., 2023), using only the first two symbols leads to an expected first null at $\frac{1}{8.8 \mu\text{s}} \approx 113 \text{ kHz}$, which is shown in the left plot of Figure 8. The narrower main lobe in the Doppler ambiguity function implies improved Doppler resolution and better acquisition and tracking performance.

5 | STARLINK SIGNAL RECEPTION

This section discusses the different signal reception setups for Starlink satellites and their associated expected received SNR and C/N_0 .

5.1 | Reception Methods

Reliable satellite communication in the Ku-band requires high-gain antennas. However, this requirement creates a challenge when considering these satellites as potential PNT sources. Specifically, high-gain antennas also have high directivity. For example, parabolic dishes and phased-array antennas, which are commonly used in this band, have a 3-dB beamwidth that ranges between sub-degrees to 5 degrees. The use of highly directive antennas imposes additional hardware and computational burdens as well as limitations on a receiver seeking to exploit these satellites for PNT. These burdens include:

- Adding the azimuth-elevation search dimensions, indicated by the antenna's pointing, to the traditional delay-Doppler dimensions during the acquisition stage (especially when the receiver is in a cold start mode and does not know

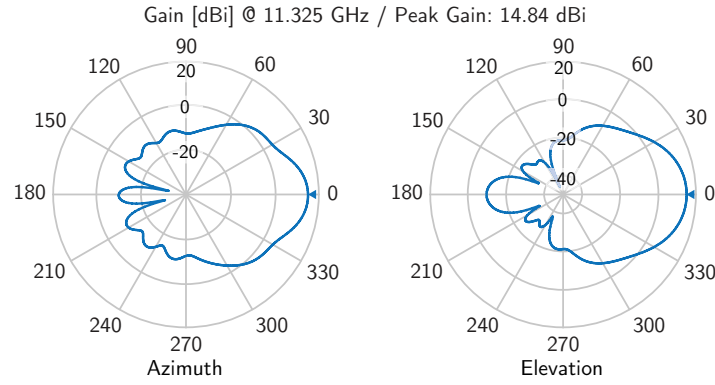


FIGURE 9 Antenna radiation pattern of the employed LNBFs in the azimuth and elevation planes.

its own position to use for an initial estimate of the overhead satellites' position).

- Constantly directing the beam towards the satellite throughout the satellite's pass to achieve a successful track.
- Tracking only one satellite at a time if the receiver has one controllable beam, which is the case with most reasonably-priced, commercially available antennas.

This paper proposes a solution to overcome these shortcomings of high-gain antennas. Specifically, this paper considers wide-beam Ku-band LNBFs to be the main RF interface with Starlink LEO satellites. Typical LNBFs have a 3-dB beam-width ranging from 15 to 30 degrees and a gain ranging from 10 to 15 dBi. This solution enables simultaneous signal reception from multiple overhead satellites and alleviates the need to constantly point towards the satellites.

However, using low-gain LNBFs alone results in low received signal power, referred to here as a low SNR regime. The low received SNR must be counteracted to ensure successful acquisition and tracking performance at the receiver. As discussed in the following sections, the low received SNR can be compensated for by maximizing the amount of signal exploited by the opportunistic blind receiver. Section 5.4 demonstrates how using the full OFDM beacon increases the processing gain at the receiver by approximately 18 dB, which both extends the effective LNBF reception pattern and counteracts the loss of gain due to using only LNBFs. The radiation pattern in the azimuth and elevation planes of the LNBFs employed in this paper are shown in Figure 9, which was generated by constructing a digital model of the employed LNBF and simulating its radiation pattern using the MATLAB Antenna Designer Toolbox (The MathWorks Inc., 2023).

5.2 | Expected C/N_0 using LNBFs

This section outlines a basic model of the power flow between a transmitter and a receiver, as illustrated in Figure 10.

Starlink LEO satellites adjust their equivalent isotropic radiated power (EIRP) density to compensate for the spreading and fading losses in the channel between the transmitter and the user equipment (Space Exploration Holdings, 2020). Users with good channels can tolerate lower nominal EIRP values, whereas users experiencing channel fading receive higher nominal EIRP values to offset the power lost

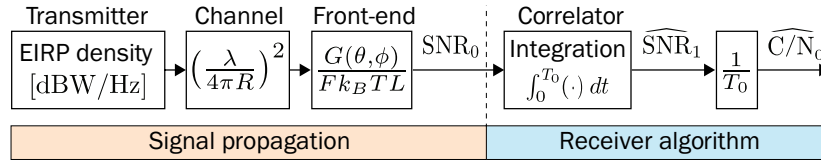


FIGURE 10 Block diagram showing a basic model of the power flow between a transmitter and a receiver, where EIRP stands for equivalent isotropic radiated power, λ is the carrier wavelength, R is the distance between the transmitter and the receiver, $G(\theta, \phi)$ is the receiver's antenna gain pattern, F is the receiver's noise figure, k_B is Boltzmann's constant, T is the system temperature, L is the aggregate system's loss, T_0 is the receiver's coherent integration time, SNR is the signal-to-noise ratio, and C/N_0 is the carrier-to-noise density ratio.

due to obstruction. Although the fast and slow fading losses cannot be predicted unless the receiver's environment is known, the free-space path loss can be readily predicted from the wavelength of the signal $\lambda \triangleq \frac{c}{f}$ and the distance R between the satellite and the user equipment.

At the receiver's RF front-end, the main factors affecting the received power are: (i) the antenna gain pattern $G(\theta, \phi)$, where $\theta \in [0, 180]$ is the elevation angle and $\phi \in [-180, 180]$ is the azimuth angle; (ii) the aggregate system's noise figure F ; (iii) the system's temperature T ; and (iv) the aggregate losses due to polarization and coupling mismatches.

At the output of the RF front-end, the SNR of the signal received from the satellite is denoted by SNR_0 . The receiver operates on the signal by match-filtering, or correlating, with a replica of the expected signal, also known as the beacon. The SNR at the input and output of the matched-filter SNR_0 and SNR_1 , respectively, are related through $\text{SNR}_1 = G_p \text{SNR}_0$, where G_p denotes the process gain of the matched filter. The SNR at the output of the matched filter is important because C/N_0 can be estimated from it. In turn, C/N_0 is a key factor for predicting the performance of the receiver's tracking loops. C/N_0 at the receiver is estimated using the signal-to-noise power ratio technique (Bhuiyan et al., 2014) given by:

$$\widehat{C/N_0}(k) [\text{dB-Hz}] \triangleq 10 \log_{10} \left\{ \frac{\widehat{\text{SNR}}_1(k)}{T_0} \right\},$$

$$\widehat{\text{SNR}}_1(k) \triangleq \frac{|Z_p(k)|^2 - |Z_n(k)|^2}{|Z_n(k)|^2}, \quad Z_p(k) \triangleq \langle \tilde{r}_k[n], s[n] \rangle, \quad Z_n(k) \triangleq \langle \tilde{r}_k[n], w[n] \rangle,$$
(5)

where $\tilde{r}_k[n]$ is the received signal after code phase and carrier phase wipe-off, as defined in Equation (4); $s[n]$ is the beacon sequence; and $w[n]$ is a noise sequence that is uncorrelated with $s[n]$ (i.e., $\langle s[n], w[n] \rangle \approx 0$) but has the same energy distribution as $s[n]$ in the time and frequency domains but a different phase distribution.

Two main Starlink beacons are currently being exploited for PNT:

1. The data-less pilot tones (Jardak & Adam, 2023; Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021). In this case, the beacon sequence is a direct-current signal $s[n] = 1, \forall n$.
2. Part of the 240 MHz OFDM signal (Grayver et al., 2024; Humphreys et al., 2023; Kozhaya et al., 2023; Neinavaie & Kassas, 2024a, 2024b). In this case, the beacon sequence is the time-domain waveform equivalent of the OFDM frame shown in Figure 5.

The following subsections discuss the expected C/N_0 with LNBs for each of the Starlink beacons.

5.2.1 | Pilot Tones

At the center of each of the eight channels in the Ku-band are nine data-less pilot tones each separated by 43,945.5 Hz. Experimental data has shown that most but not all Starlink satellites have these tones active as shown in Data set D1 Section 8.2.1. Furthermore, experimental data collected by the authors from March 2021 through January 2024 revealed that the peak C/N_0 per tone decreased from 50 dB-Hz in the initial years (before 2023) to 20 dB-Hz afterwards. While the high peak C/N_0 during the initial years enabled accurate carrier and Doppler tracking with LNBs (Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021); the current peak C/N_0 , which is lower than the threshold for reliable tracking, rendered the LNBs practically unusable.

5.2.2 | OFDM Signals

According to filings submitted by Starlink to the Federal Communication Commission (Space Exploration Holdings, 2020), the nominal EIRP density at nadir is -56.22 dBW/Hz and the nominal distance R to the receiver is 550 km. Considering the channel at 11.325 GHz, the corresponding wavelength λ is 2.65 cm and the peak LNB gain is $G(0, 0) = 14.84$ dBi. The aggregate system's noise figure is calculated to be $F = 1.1$ dB (of which 0.8 dB is due to the LNB and the remainder is insertion loss). The nominal system's temperature is considered to be $T = 290$ K, and the loss due to receiving a circularly-polarized signal with linearly-polarized LNB is $L = 3$ dB.

Using these nominal values in the received power model leads to a maximum expected SNR of:

$$SNR_0 = EIRP \frac{\lambda^2 G(0, 0)}{(4\pi R)^2 F k_B T L} = -9.83 \text{ dB}.$$

The maximum C/N_0 is given by:

$$C/N_0 [\text{dB-Hz}] = SNR_0 [\text{dB}] + 10 \log_{10}(B), \quad (6)$$

where B is the receiver bandwidth in Hz. Figure 11 shows the maximum achievable C/N_0 using LNBs as a function of the receiver's bandwidth. Note that the C/N_0 does not increase for $B \geq 240$ MHz because the bandwidth of the OFDM beacon is 240 MHz.

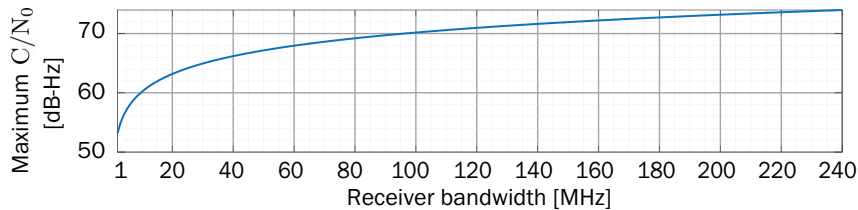


FIGURE 11 Maximum achievable C/N_0 versus receiver bandwidth when receiving OFDM signals from a Starlink satellite using typical LNBs.

5.3 | The Starlink Ku-Band Spectrum

When a Starlink Ku-band downlink channel is active, it can exhibit either one or both of the pilot tones and OFDM beacons. The left plot in Figure 12 shows a full bandwidth capture of one Starlink Ku-band downlink channel, while the right plot of Figure 12 shows Starlink's OFDM and pilot tone beacons as received by a low-gain and a high-gain setup before and after the year 2023. Note that the same LNBF was used in the recordings before and after 2023, and the power levels in the spectrum shown in Figure 12 were adjusted to match the OFDM power matched between the two. Comparing the recordings before and after 2023 indicates that the spectrum of the pilot tones received by an LNBF before 2023 was approximately 30 dB lower, making a high-gain antenna (e.g., a 30 dBi dish) crucial for receiving the pilot tones. In addition, note that the power level of the pilot tones shown in Figure 12 is a sample realization for a specific satellite. The power of the pilot tones is also not consistent throughout the satellites; it seems to depend on many parameters, including both the receiver's and satellite's antenna pointing.

5.4 | Processing Gain

The processing gain G_p of the matched-filter is equivalent to the time-bandwidth product of the signal (i.e., beacon) used in the correlation (Mahafza, 2016). For instance, in GPS, which employs a direct-sequence spread spectrum modulation technique, the pseudorandom noise sequence in the L1 C/A channel has a bandwidth $B_e = 1.023$ MHz and spans a duration of $T_e = 1$ ms. The processing gain due to correlating with the known 1 ms pseudorandom noise is $G_p = B_e T_e = 30$ dB. Moreover, after bit synchronization, the receiver can increase the coherent integration time up to 20 ms, leading to a processing gain of $G_p = 20 B_e T_e = 43.1$ dB. This gain is what enables the GPS receiver to pick up the satellite's signal from the noise floor despite the low received SNR.

In a similar fashion to GPS, this paper demonstrates how exploiting the full OFDM beacon of Starlink satellites can provide enough processing gain for the receiver to attain the maximum achievable C/N_0 , enabling practical exploitation of received signals for PNT in a low SNR regime with commercial off-the-shelf LNBFs.

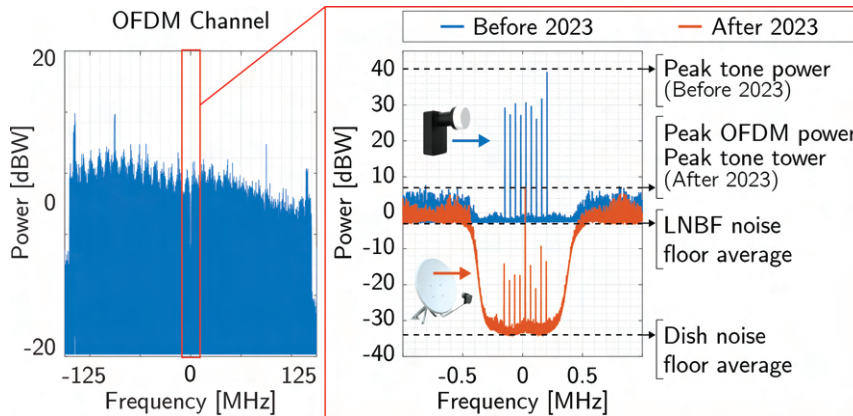


FIGURE 12 Starlink's OFDM and pilot tones beacons as received by an LNBF (low-gain) and a dish (high-gain) setup. *Left:* A capture of one of the 250 MHz Ku-band channels using a 35 dBi dish. *Right:* Zoomed-in view of the nine pilot tones present in the 1 MHz silent band at the center of the channel. The zoom shows a snapshot of Starlink's pilot tones using an LNBF before 2023 (in blue) and a 35 dBi dish after 2023 (in orange).

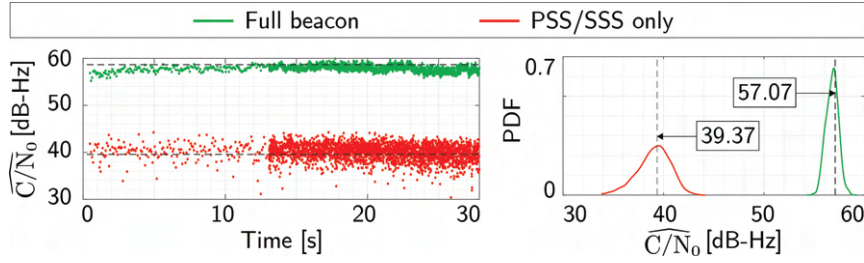


FIGURE 13 Empirical $\widehat{C/N_0}$ for a 5 MHz LNBF receiver correlating with the full beacon (green) versus PSS/SSS only (red). *Left:* $\widehat{C/N_0}$ versus time. *Right:* the empirical probability density function of the received $\widehat{C/N_0}$.

Substituting $\text{SNR}_1 = G_p \text{SNR}_0$ in (5) yields:

$$\mathbb{E}\left\{\widehat{C/N_0}(k)\right\} = 10 \log_{10} \left\{ \frac{G_p}{T_0} \text{SNR}_0(k) \right\}. \quad (7)$$

Note that, as illustrated in Figure 10, the SNR at the input of the receiver SNR_0 does not depend on the receiver design being used. Nonetheless, for two different receivers correlating the incoming signal with two different beacons, the SNR_1 at the output of their correlators, and in turn the $\widehat{C/N_0}$, will be different due to the different values of G_p for each beacon. The more knowledge the receiver has about the incoming signal (i.e., using the full beacon), the greater the value of G_p , SNR_1 , and $\widehat{C/N_0}$. Note that, for a receiver with bandwidth B and integrating time T_0 , the relationship $0 \leq G_p \leq BT_0$ cannot be violated. In the limit, when the beacon spans the whole time-bandwidth resource, $G_p = BT_0$, and Equation (7) tends to Equation (6).

The gain due to match-filtering with the blindly estimated beacon was numerically evaluated to be $G_p = 38.2$ dB for $B = 5$ MHz, leading to $\mathbb{E}\left\{\widehat{C/N_0}\right\} = 57.07$ dB-Hz when the satellite is transmitting at its highest power. By comparison, using only the first two symbols of the OFDM frame (i.e., PSS and SSS; Humphreys et al., 2023), leads to a processing gain of $G_p = 20.5$ dB for $B = 5$ MHz and $\mathbb{E}\left\{\widehat{C/N_0}\right\} = 39.37$ dB-Hz.

It was observed experimentally that a C/N_0 value of $40 + \text{dB-Hz}$ is needed for reliable acquisition and tracking. Maximizing the processing gain by exploiting the full beacon through blind estimation therefore enables successful receiver operation in low SNR regimes. Figure 13 shows an example of a Starlink satellite tracked using only PSS/SSS vs. the full beacon. The left plot shows the $\widehat{C/N_0}$ for a 5 MHz LNBF receiver correlating with the full beacon (in green) versus the PSS/SSS (in red) (Humphreys et al., 2023); the right plot shows the empirical probability density function (pdf) of the received $\widehat{C/N_0}$ for both beacons, which closely matches the theoretically predicted values discussed above. Some minor discrepancy is caused by the imperfect modeling of the experimental setup.

6 | STARLINK PNT RECEIVER DESIGN

This section describes the receiver design needed to exploit Starlink LEO signals for PNT. The Starlink receiver, similar to traditional GNSS receivers, operates in two stages: acquisition and tracking.

In the acquisition stage, the receiver uses the beacon shown in Figure 5 and cross-correlates it with the incoming signal. For every T_0 seconds, the receiver searches over a predefined set of delay and Doppler frequencies and checks if the Starlink beacon exists in the incoming signal. For this paper, the Doppler search range was -250 kHz to 250 kHz with a bin size of 100 Hz, and the delay search range was set between $-\frac{T_0}{2T_s}$ s and $\frac{T_0}{2T_s}$ s with a bin size of T_s s.

When a Starlink beacon is acquired, the receiver proceeds to track the satellites as described below. For tracking, define the state vector $\mathbf{x}(t) \triangleq [\theta(t), \tau(t)]^\top$, where $\theta(t) \triangleq [\theta(t), \dot{\theta}(t), \ddot{\theta}(t)]^\top$ is the beat carrier phase state vector and $\tau(t) \triangleq [\tau(t), \dot{\tau}(t)]^\top$ is the code phase state vector.

The dynamical model of $\mathbf{x}(t)$ is given by:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\tilde{\mathbf{w}}(t),$$

$$\mathbf{A} \triangleq \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{B} \triangleq \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \tilde{\mathbf{w}} \triangleq \begin{bmatrix} \tilde{w}_\theta \\ \tilde{w}_\tau \end{bmatrix}, \quad (8)$$

where $\tilde{w}_\theta(t)$ and $\tilde{w}_\tau(t)$ are zero-mean white noise processes with power spectral density $\tilde{q}_\theta$ and $\tilde{q}_\tau$, respectively. In this paper, $\tilde{q}_\theta$ and $\tilde{q}_\tau$ were tuned manually to $(2\pi)^2 \text{ rad}^2/\text{s}^3$ and $(0.004)^2 \text{ s}^2/\text{s}$, respectively. Note that, while this paper does not account for the Doppler stretch for reasons discussed in Section 3, the Kalman filter (KF) is still designed to track the first-order term of the code phase. This design serves mainly to enhance the stability of the KF's code phase prediction in low pulse repetition frequency (PRF) regimes (where the measurement update rate is low).

The continuous-time model given in Equation (8) is discretized at a constant sampling time T_0 , leading to:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k,$$

where $\mathbf{x}_k \triangleq [\theta_k, \tau_k]^\top$, $\theta_k \triangleq [\theta_k, \dot{\theta}_k, \ddot{\theta}_k]^\top$ is the beat carrier phase state vector at the k -th accumulation, $\tau_k \triangleq [\tau_k, \dot{\tau}_k]^\top$ is the code phase state vector at the k -th accumulation, $\mathbf{F} \triangleq e^{\mathbf{A}T_0}$ is the state transition matrix, and $\mathbf{w}_k$ is a discrete-time zero-mean white random sequence with covariance $\mathbf{Q} = \int_0^{T_0} e^{\mathbf{A}t} \mathbf{B} \tilde{\mathbf{Q}} (e^{\mathbf{A}t} \mathbf{B})^\top dt$, where $\tilde{\mathbf{Q}} \triangleq \begin{bmatrix} \tilde{q}_\theta & 0 \\ 0 & \tilde{q}_\tau \end{bmatrix}$.

The reconstructed beat carrier phase sequences that are used to perform wipe-off in Equation (4) can be written as $\hat{\theta}_k[n] = \hat{\theta}_k + \hat{\dot{\theta}}_k n T_s + \frac{1}{2} \hat{\ddot{\theta}}_k (n T_s)^2$. After performing code and carrier wipe-off, given in Equation (4), the receiver correlates the received accumulation with a bank of matched-filters whose output is fed to discriminators that quantify the beat carrier phase, code phase, and frequency error at the k -th accumulation. The matched-filters' bank is given by:

$$Z_P(k) \triangleq \langle \tilde{r}_k[n], s[n] \rangle, \quad Z_\beta^\alpha(k) \triangleq \langle \tilde{r}_k[n], s_\beta^\alpha[n] \rangle,$$

where $\alpha \in \{t, f\}$, $\beta \in \{E, L\}$, and

$$s_E^t[n] \triangleq (s \otimes -\delta_t)[n], \quad s_E^f[n] \triangleq s[n] \exp[-j2\pi\delta_f n T_s],$$

$$s_L^t[n] \triangleq (s \otimes +\delta_t)[n], \quad s_L^f[n] \triangleq s[n] \exp[+j2\pi\delta_f n T_s],$$

where $\delta_t = 0.2 \mu s$ and $\delta_f = 350 \text{ Hz}$, thereby representing half of the early-minus-late spacing in the time- and frequency-domains, respectively. The output of the correlators is used to construct the error vector $\mathbf{v}_k \triangleq [\tilde{\theta}_k, \tilde{\theta}_k, \tilde{\tau}_k]^\top$, with:

$$\tilde{\theta}_k = \text{atan2}(\Im\{Z_P(k)\}, \Re\{Z_P(k)\}), \quad \tilde{\theta}_k = \pi \frac{|Z_E^f(k)| - |Z_L^f(k)|}{|Z_E^f(k)| + |Z_L^f(k)|}, \quad \tilde{\tau}_k = \frac{1}{2} \frac{|Z_E^t(k)| - |Z_L^t(k)|}{|Z_E^t(k)| + |Z_L^t(k)|},$$

where $\Re\{Z\}$ and $\Im\{Z\}$ denote the real and imaginary parts of the complex number Z .

Finally, the error vector is fed as an innovation to a KF loop that uses the observation model $\mathbf{z}_k = \mathbf{C}\mathbf{x}_k + \mathbf{v}_k$, where:

$$\mathbf{C} \triangleq \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{v}_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \sigma_\theta^2 & 0 & 0 \\ 0 & \sigma_f^2 & 0 \\ 0 & 0 & \sigma_\tau^2 \end{bmatrix}\right).$$

In this paper, σ_θ, σ_f , and σ_τ were tuned manually to $(2\pi \times 10^{-5}) \text{ rad}$, $(\pi \times 10^{-2}) \text{ rad/s}$ and $(3 \times 10^{-7}) \text{ s}$, respectively.

7 | STARLINK'S OBSERVABLES

This section presents a detailed analysis of the quality of navigation observables that can be extracted from Starlink satellites. The plots shown in the following sections are based on data sets collected in different years (2021 to 2024) and locations (Irvine, CA, USA and Columbus, OH, USA). Table 2 shows the parameters for the three data sets that were used to develop the subsequent sections. Note that the center frequencies in all data sets were active during data collection.

7.1 | Pulse Repetition Frequency

Starlink satellites transmit user data in OFDM frames of duration T_0 . While the pilot tones, if they are present at the center of the channel, remain active during

TABLE 2
List of Data Set Collection Parameters

Parameter	D1	D2	D3
Location	Irvine, CA, USA	Columbus, OH, USA	Columbus, OH, USA
Date	March 2021	December 2023	July 2024
Antenna	LNBF only	LNBF only	LNBF only
Bandwidth [MHz]	2.5	5	2.5
Frequency [GHz]	11.325	11.075, 11.325, 11.575	10.825, 11.075, 11.325, 11.575, 11.825, 12.075, 12.325, 12.575
Duration [min]	15	45	10
Number of satellites	8	83	63
Exploited beacons	Pilot tones and OFDM	OFDM	OFDM

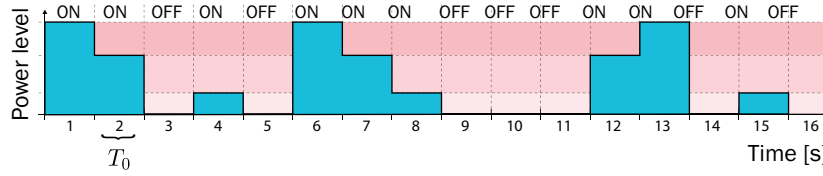


FIGURE 14 Sketch showing the transmission scheme of Starlink satellites. Different shades of red illustrate different clusters of transmitted power levels. The transmitted (ON) frames of duration T_0 seconds are highlighted in blue, while the OFF frames have an OFDM power level of zero.

the entire transmission period, the availability of the OFDM signal depends on the users' activity in the serviced cell.

This paper refers to the ratio of transmitted frames (i.e., frames with an OFDM signal) to the total number of frames spanning a given period of time as the PRF. For instance, Figure 14 shows nine transmitted (ON) OFDM frames and 7 that are OFF, so the PRF during this transmission period is $\frac{9}{9+7} \approx 0.56$. The PRF can range from 0, indicating no transmission, to 1, indicating continuous transmission.

Previous studies evaluated over half a billion packet timestamps at the user terminal and attempted to infer Starlink's physical layer transmission rates (Garcia et al., 2023; Garcia et al., 2024). These studies concluded the following:

- **P1:** There are 14 likely physical layer rates. The baseline rate is 430.5 Mbps, and higher rates are greater than the baseline by multiples of 27 Mbps.
- **P2:** There is a short silent reconfiguration interval that occurs every 15 seconds when the physical layer rate (referred to as the transmission mode in this paper) changes.

Figure 15(a) shows the C/N_0 and PRF from a 75-second tracking of a Starlink satellite (Data set D1, NORAD ID 45693), during which consistent and simultaneous changes in the C/N_0 and PRF curves were detected every 15 seconds. The dip in the PRF every 15 seconds matches observation P2, indicating a transmission mode change.

Because multiple users are serviced by the same satellite, the PRF for a given Starlink satellite at a given channel is technically proportional to the sum of the physical layer rates from all active users'. Observation P1 indicates that the physical layer rate does not fall below baseline (430.5 Mbps) for an *active* Starlink satellite servicing an *active* cell. This means that the tracked PRF for a given *active* Starlink satellite should always be greater than a threshold, referred to as the baseline PRF in this paper. For active cells with low user activity, the expected PRF is low, albeit always greater than a baseline which was estimated from the empirical data to be around 2%. On the other hand, active cells with high user activity present higher values of PRF, reaching 100% most of the time.

Figure 15(b) shows the empirical probability density function of the PRF based on data sets D1 and D2 from Table 2. The green curve, which corresponds to D1, showed PRF values closer to unity and will therefore be referred to as a high-activity cell. The red curve, which corresponds to D2, showed low PRF values and will therefore be referred to as a low-activity cell. In agreement with observation P1, the plots show the baseline PRF as well as the higher modes that were observed when multiple users and/or higher rates were being serviced.

Based on the above information, the maximum achievable OFDM-based measurement rate from an *active* Starlink satellite can be inferred to be T_0^{-1} . $100\% = 750 \text{ Hz}$;

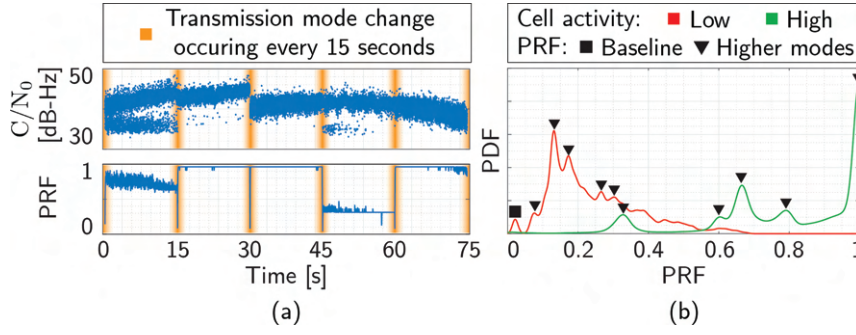


FIGURE 15 Effect of Starlink subscriber activity on signal availability. (a) A 75-second sample track of a Starlink satellite exhibiting consistent change in the transmission mode every 15 seconds. (b) An empirical probability density function of the pulse repetition frequency in low activity regimes (red) and high activity regimes (green).

this is mostly the case when servicing high-activity cells. Moreover, the minimum (baseline) OFDM-based measurement rate from an *active* Starlink satellite is $T_0^{-1} \cdot 2\% = 15$ Hz; this is mostly the case when servicing low-activity cells.

7.2 | Doppler Shift

Before year 2024, the OFDM-based Doppler tracking exhibited consistent and sudden CFO corrections occurring every second. These CFO corrections are alternatively referred to as Doppler bias in this paper. While previous research showed precise carrier/Doppler tracking and positioning with Starlink satellites (Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021), these papers mainly exploited the data-less pilot tones, which, unlike the OFDM-based signals, are not contaminated by CFO corrections.

Figure 16(a) shows the OFDM-based Doppler bias present in one of the tracked Starlink satellites from D1. The black curve shows the difference between the tracked pilot tone-based Doppler and the predicted Doppler calculated using the receiver and satellite positions, which were obtained using data from two-line element (TLE) files and propagated with simplified general perturbations 4 (SGP4) orbit propagation. The blue curve shows the difference between the tracked OFDM-based Doppler and the predicted Doppler calculated using TLE+SGP4 orbit propagation. The orange curve shows the receiver's attempt to estimate the Doppler bias on-the-fly. Note that the TLE+SGP4 ephemeris is assumed to be the true position of the satellite after correcting the temporal and orbital errors using knowledge of the receiver's position (Hayek et al., 2024).

The Doppler bias estimation framework employed here works by incorporating prior knowledge of this abrupt, every-second jump in the tracked Doppler into its tracking loop. In the time window where a CFO correction is expected, the receiver estimates this jump by subtracting the open-loop KF-predicted Doppler shift from the closed-loop KF-estimated Doppler shift. Recall from Section 6 that $\hat{\theta}_k \triangleq [\hat{\theta}_k, \hat{\theta}_k, \hat{\theta}_k]^T$ is the estimated beat carrier phase state vector at the k -th accumulation. Use $\hat{\theta}_k^0 \triangleq [\hat{\theta}_k^0, \hat{\theta}_k^0, \hat{\theta}_k^0]^T$ to denote the modified estimated beat carrier phase state vector given by:

$$\hat{\theta}_k^0 \triangleq \begin{cases} \hat{\theta}_k, & \Delta_M \leq (k + k_0) \bmod M \leq M - \Delta_M, \\ \mathbf{F}_\theta \hat{\theta}_{k-1}^0, & \text{otherwise,} \end{cases}$$

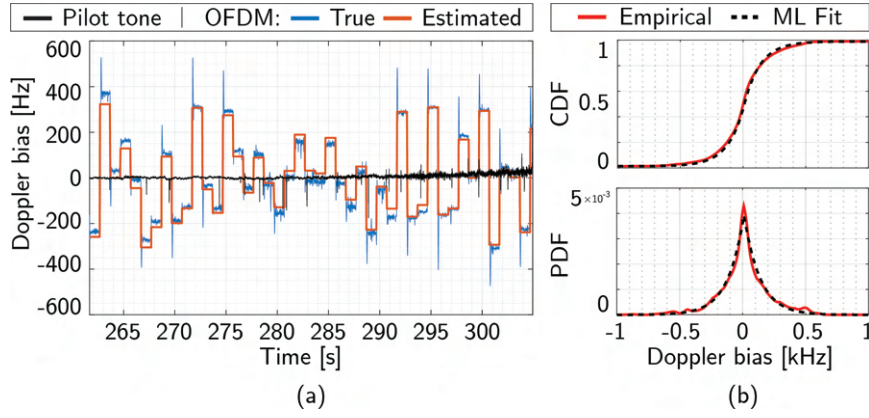


FIGURE 16 Pilot tone versus OFDM-based Doppler tracking with Starlink signals before year 2024. (a) *Black*: difference between the tracked pilot tone-based Doppler and the expected Doppler calculated from TLE+SGP4 orbit propagation. *Blue*: difference between the tracked OFDM-based Doppler and the expected Doppler calculated from TLE+SGP4 orbit propagation. *Orange*: estimated Doppler bias within the proposed receiver framework. (b) Empirical cumulative and probability distribution functions of the OFDM-based Doppler bias (solid line) and its maximum likelihood Laplacian fit (dashed line).

where $k_0 \in \{0, \dots, M-1\}$ is a constant that aligns the receiver's time to the one-second time grid of the CFO correction; $M \in \mathbb{N}$ is the number of accumulations per second, which is $M = 750$ for Starlink; $\Delta_M \in \{1, \dots, \lfloor \frac{M}{2} \rfloor\}$ is the half-width of the Doppler jump estimation window (chosen to be 50 in this paper); $x \bmod y$

is the modulo operation; and $\mathbf{F}_\theta \triangleq \begin{bmatrix} 1 & T_0 & \frac{T_0^2}{2} \\ 0 & 1 & T_0 \\ 0 & 0 & 1 \end{bmatrix}$. The integer k_0 is estimated at

the receiver by examining multiple consecutive one-second intervals and selecting the modulo index at which the Doppler innovation repeatedly exhibits an abrupt impulse.

Finally the CFO at the k -th accumulation is retrieved as $\text{CFO}_k = \sum_{j=0}^k \hat{\theta}_j^e$, where $\hat{\theta}_k^e$ is the Doppler error signal given by:

$$\hat{\theta}_k^e \triangleq \begin{cases} \frac{1}{\Delta_M} \sum_{j=k}^{k+\Delta_M-1} (\hat{\theta}_j - \hat{\theta}_j^0), & (k+k_0) \bmod M = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Figure 16(b) shows, with solid red lines, the empirical cumulative distribution function and pdf of the OFDM-based Doppler biases calculated from data sets D1 and D2. A maximum likelihood estimator (Norton, 1984) was adopted to calculate the parameters of the empirical pdf, and it showed that the pdf follows approximately a Laplacian distribution with a mean of zero and standard deviation of 180 Hz.

The OFDM signals in the data are not contaminated with CFO corrections as of year 2024. Figure 17 shows the error between the tracked OFDM-based Doppler and the Doppler predicted from TLE+SGP4 propagation. Figure 17(a) shows a sample of the Doppler while Figure 17(b) shows its empirical distribution calculated from data set D3. Data fitting shows that the error can be appropriately modeled as a zero-mean Gaussian distribution with a standard deviation of 30 Hz. This error is assumed to be induced mainly by the tracking loops and apparatus noise.

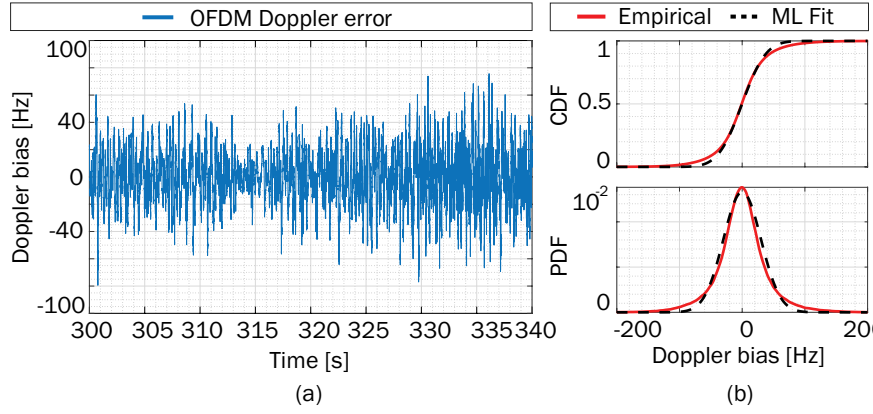


FIGURE 17 OFDM-based Doppler tracking with Starlink signals as of year 2024. (a) Difference between the tracked OFDM-based Doppler and the expected Doppler calculated from TLE+SGP4 orbit propagation. (b) The empirical cumulative (cdf) and probability density functions (pdf) of the Doppler error (solid lines) and their maximum likelihood Gaussian fit (dashed line).

7.3 | Carrier Phase

Due to the abrupt CFO corrections discussed above, the OFDM-based carrier phase exhibited excessive cycle slips every second before year 2024. However, this behavior is no longer recorded as of year 2024. On the other hand, a closer inspection of the C/N_0 , carrier phase error, and angle of the prompt correlator revealed that different clusters are readily identifiable through different power allocations (measured by the C/N_0) and phase offsets. Similar to previous observations (Humphreys et al., 2023), these clusters are likely related to different Starlink users. In this paper, these clusters are therefore referred to as users.

Figure 18 shows the C/N_0 , carrier phase error, and angle of the prompt correlator in the IQ plot during the track of a Starlink satellite. In Figure 18, the receiver is mainly locked on the carrier phase of user 2 (shown in blue), which has the highest PRF compared to the other two users (users 1 and 3). When the satellite is transmitting a frame to user 1 or 3, the carrier phase error exhibits an abrupt jump of $\frac{\pi}{4}$ or $\frac{\pi}{2}$, and the power at the correlator changes depending on the power allocated for the given user. The IQ plot at the right of Figure 18 suggests that each user has a specific power and phase reference. The bottom plot of Figure 18 shows a close-up of the carrier phase error:

1. The blue curve at the bottom of Figure 18 is associated with the OFDM signal and is mainly driven by the user with the highest PRF (user 2). The other two users introduce excessive $\frac{\pi}{4}$ or $\frac{\pi}{2}$ slips whenever the frame is directed to them.
2. The red curve is associated with the pilot tones. This curve is barely noticeable because the pilot tones are not contaminated by the modulation and multiple-access schemes and therefore remain near zero.

7.4 | Code Phase

OFDM-based code phase tracking with Starlink signals exhibits two main types of corrections. The first, referred to as a macro-correction, occurs every 15 seconds and coincides with the transmission mode change discussed in Section 7.1. These

macro-corrections take values ranging from nanoseconds to milliseconds. During the macro-correction, the code phase dynamics are drastically altered, to the point where they do not match predictions based on knowledge of the receiver's location and satellite's position through orbit propagation. While the receiver can estimate and track the CFO corrections in the OFDM-based Doppler with good fidelity, as shown in Figure 19, on-the-fly estimation of the macro-corrections is impossible

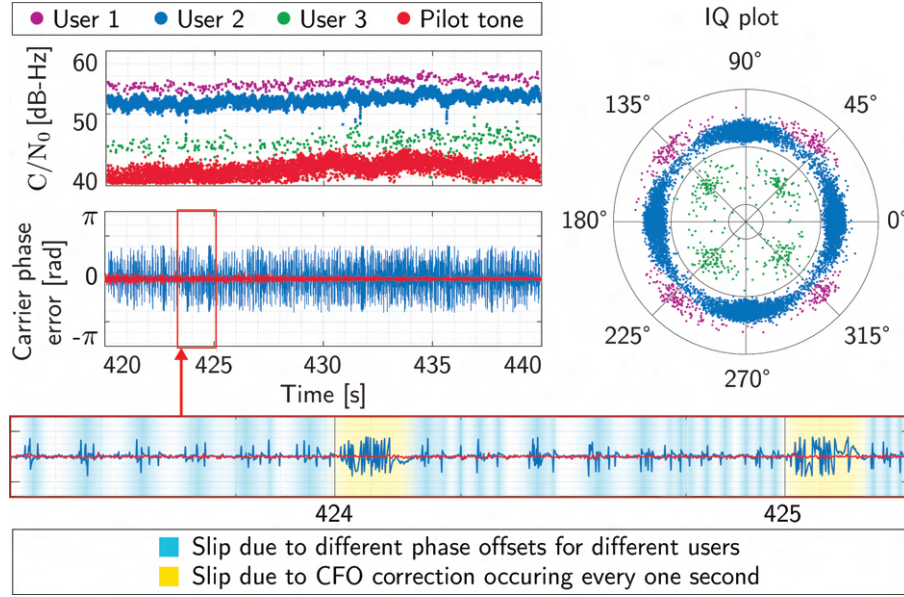


FIGURE 18 OFDM-based versus pilot tone-based carrier phase tracking of Starlink signals. *Top:* C/N_0 curves of: (i) the OFDM signal, showing three clusters corresponding to three different users (purple, blue, and green), and (ii) the non-OFDM, data-less, pilot tone signal (red). *Right:* IQ plot of the prompt correlator showing different power levels and phase offsets for the three different users. *Bottom:* Zoomed-in view of the carrier phase error showing: (i) the OFDM-based carrier phase, with recurrent phase slips due to CFO correction and the multiple access scheme, and (ii) the pilot tone-based carrier phase, which has no slips because it is unmodulated.

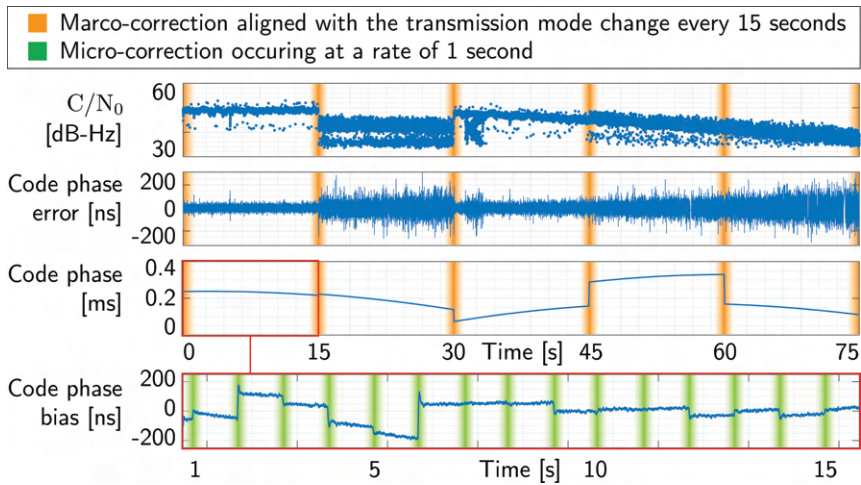


FIGURE 19 OFDM-based code phase tracking with Starlink signals. From top to bottom: (i) the C/N_0 curve of a sample tracked satellite; (ii) the code phase error during the tracking period; (iii) the tracked code phase exhibiting abrupt macro-corrections in the dynamics, which are aligned with the transmission mode change occurring every 15 seconds; and (iv) zoom of the first 15-second interval, showing nanosecond-level micro-corrections occurring every second.

due to the alteration of not only the zeroth-order term of the code phase (bias) but also the higher-order terms.

The second type of correction are micro-corrections, which occur once every second and take values in the range of nanoseconds. Figure 19 shows the C/N_0 , code phase error, and code phase of a tracked Starlink satellite (Data set D1, NORAD ID 46572). The code phase shows the macro-corrections coinciding with the transmission mode change occurring every 15 seconds. Moreover, a zoom of the first 15-second interval of the code phase shows the nanosecond-level micro-corrections occurring every second.

The OFDM-based code phase is not contaminated with micro-corrections as of year 2024, but the macro-corrections are still observed.

8 | POSITIONING WITH STARLINK SIGNALS

This section presents the first published results for successful positioning using the full Starlink OFDM signals in a low SNR regime imposed by the use of low-gain LNBFs. Three experimental positioning results are presented based on data sets D1, D2, and D3 (Table 2). The results for data sets D1 and D2 assess the performance of opportunistic positioning using Starlink's OFDM signals during the initial phase of the constellation deployment, and the results for data set D3 assess the latest positioning performance given the current status of the constellation.

8.1 | Positioning Filter

Let $i \in [1, L]$ denote the Starlink LEO satellite's index, where L represents the total number of tracked satellites. The Doppler shift measurements $f_{D,i}(k)$, in Hz, extracted from the i -th LEO satellite at time-steps k as discussed in Section 3, are converted to pseudorange rate measurements $\dot{\rho}_i(k)$, expressed in m/s, using the relationship $\dot{\rho}_i(k) = -c \frac{f_{D,i}(k)}{f_c}$. The pseudorange rate can be modeled as:

$$\dot{\rho}_i(k) = -\dot{\mathbf{r}}_{SV,i}(k')^\top \frac{\mathbf{r}_r(k) - \mathbf{r}_{SV,i}(k')}{\|\mathbf{r}_r(k) - \mathbf{r}_{SV,i}(k')\|_2} + c[\dot{\delta}t_{clk,i}(k) + \dot{\delta}t_{atm,i}(k)] + v_i(k) \quad (9)$$

where $\mathbf{r}_r$ and $\mathbf{r}_{SV,i}$ are the 3D position vectors of the receiver and the i -th satellite, respectively, in the east-north-up (ENU) reference frame; $\dot{\mathbf{r}}_{SV,i}$ is the 3D velocity vector of the i -th satellite in the ENU frame; $\dot{\delta}t_{clk,i}$ is the lumped clock drift of the receiver and the i -th satellite; $\dot{\delta}t_{atm,i}$ is the lumped ionospheric and tropospheric delay rates; and v_i is the pseudorange rate measurement noise, which is modeled as a discrete-time, zero-mean white Gaussian sequence with variance $\sigma_{v_i}^2(k)$. In (9), k' represents the discrete-time instance at $t_{k'} = t_k - \delta t_{TOF_i}$, with δt_{TOF_i} being the true time-of-flight (TOF) of the signal from the i -th LEO satellite. This paper assumes $k' \approx k$ to simplify the formulation of the positioning filter. Although this approximation creates an error in the satellites' position, this error is negligible compared to the ephemeris error in the TLE, from which the SGP4-propagated ephemeris is obtained. The error introduced by the $k' \approx k$ approximation is lumped into a combined term describing the delay between the receiver and each satellite, and then this term is estimated as described below. The ionospheric attenuation is inversely proportional to the square-root of the carrier frequency, and hence can be neglected for Ku-band signals. Similarly, the tropospheric delay rate is ignored in this study, leading to $\dot{\delta}t_{atm,i}(k) = 0$.

The state vector is given by:

$$\mathbf{x} = [\mathbf{r}_r^T, \mathbf{x}_{\text{clk},1}^T, \dots, \mathbf{x}_{\text{clk},L}^T]^T, \quad \mathbf{x}_{\text{clk},i} \triangleq [c\delta t_{\text{clk},i}, c\dot{\delta} t_{\text{clk},i}]^T.$$

The Starlink OFDM beacon, shown in Figure 5, can be approximated by a model beacon having a uniform power level in the interval $[0, T_e]$, where T_e is the effective beacon length in seconds (Kozhaya & Kassas, 2024b). According to (Kozhaya & Kassas, 2024b), Starlink's OFDM beacon has an effective length of $T_e \approx 1$ ms. Furthermore, because the assumed uniform power level in the interval $[0, T_e]$ is similar to that of the GPS PRN, which is $|\pm 1|^2 = 1$, the C/N_0 and measurement noise can be related to each other given the relationships developed in (Kaplan & Hegarty, 2005). The received C/N_0 , which is estimated by the receiver as discussed in Section 6, was therefore used to calculate the weight associated for each measurement in the weighted batch nonlinear least squares (WNLS) estimator. The weight for the k -th measurement, in $(\text{m/s})^2$, is given by:

$$\sigma_{v_i}^2(k) = \left[\frac{c\delta_f \sigma_s}{f_c \pi T_e} \right]^2 \frac{1}{(C/N_0)_i(k)} \left[1 + \frac{1}{T_e (C/N_0)_i(k)} \right],$$

where $\sigma_s = 7.5$ is a unit-less inflation factor to account for other unmodeled errors.

A total of 200 Monte Carlo (MC) realizations were generated for each scenario by randomizing the initial receiver's position estimate in the ENU local navigation frame according to the distribution $\mathbf{x}_r(0) \sim \mathcal{N}(\mathbf{0}_{3 \times 1}, \mathbf{P}_r)$, where $\mathbf{P}_r = \text{diag}[10^{10}, 10^{10}, 10^4]$ with units of $[\text{m}^2, \text{m}^2, \text{m}^2]$. The initial clock bias and drift terms for each satellite were initialized to zero. The satellites' ephemerides were obtained by propagating the corresponding TLE files via SGP4 at the time of the experiment. Note that the TLE+SGP4 ephemeris is assumed to be the true position of the satellite after correcting the temporal and orbital errors using knowledge of the receiver's position (Hayek et al., 2024).

8.2 | Data Sets 1 and 2

The first data set (D1) was collected in March 2021 on the roof of the Anteater parking structure at the University of California, Irvine, CA, USA. During that time, the pilot tones were reliably detectable and trackable by LNBs. Both the pilot tones and OFDM-based beacons were exploited for this data set. In this data set, the OFDM-based Doppler was contaminated by Doppler biases. To compensate for this, four different frameworks for positioning with Starlink LEO satellites were developed and compared:

- **F1:** Pilot tone-based Doppler shift tracking that exhibits no sign of contamination from OFDM-related corrections.
- **F2:** OFDM-based Doppler shift with uncorrected CFOs.
- **F3:** OFDM-based Doppler shift with corrected CFOs that are estimated on-the-fly by the receiver's tracking loop.
- **F4:** OFDM-based Doppler shift with corrected CFOs that are estimated using an assumed cooperative base station that is receiving signals from the same Starlink satellites. This base station estimates the Doppler bias history by comparing the tracked Doppler shift with its expected value calculated from knowledge of the base station's position and the satellite's ephemeris using TLE+SGP4 orbit propagation.

The second data set (D2) was collected in December 2023 on the roof of the ElectroScience Laboratory at The Ohio State University, Columbus, OH, USA. At this time, the pilot tones were not reliably detectable and trackable by LNBs, so only the OFDM beacon is exploited for this data set. As with D1, the OFDM-based Doppler in D2 was contaminated by Doppler biases, so the precision of frameworks F2, F3, and F4 were compared for this data set as well.

8.2.1 | Acquisition and Tracking

In the first data set, a total of eight Starlink satellites were tracked for 15 minutes. The hardware setup consisted of an upward-pointing LNB connected to a USRP 2974 with a sampling rate of 2.5 MHz at a carrier frequency of 11.325 GHz. The left plots of Figure 20 show the acquisition plot for one of the acquired Starlink satellites as well as the skyplot of the eight tracked satellites.

In the second data set, a total of six Starlink satellites were tracked during the course of eight minutes. The hardware setup consisted of an upward-pointing LNB connected to a USRP 2955 with a sampling rate of 5 MHz at carrier frequencies of 11.075, 11.325, and 11.575 GHz. The right plots of Figure 20 show the acquisition plot for one of the acquired Starlink satellites as well as the skyplot of the six tracked satellites.

After acquisition, the Starlink satellites were tracked using both the pilot tones (in D1) and OFDM beacons (in D1 and D2). Figure 21 summarizes the tracking results for D1 on the left and D2 on the right. Figure 21(a) shows the estimated C/N_0 curves of the pilot tones and OFDM beacons in black and blue, respectively. The PRF of the downlink during the active OFDM transmission regions is plotted in Figure 21(b), which shows high user activity in D1 and lower activity in D2. This result implies high measurement update rates in D1 and lower rates in D2. Figure 21(c) plots the Doppler shift of the pilot tones and OFDM beacons in black and dashed blue, respectively. Figure 21(d) plots the OFDM-based code phase, in which the macro-corrections are clearly observable. Figure 21(e) plots an overview of the error between the tracked Doppler and the Doppler calculated using knowledge of the receiver's and satellite's positions. The error curves correspond to the four different frameworks discussed above: (i) pilot tones-based Doppler in black, (ii) OFDM-based Doppler without corrections in blue, (iii) OFDM-based Doppler with on-the-fly correction in orange, and (iv) OFDM-based Doppler with correction

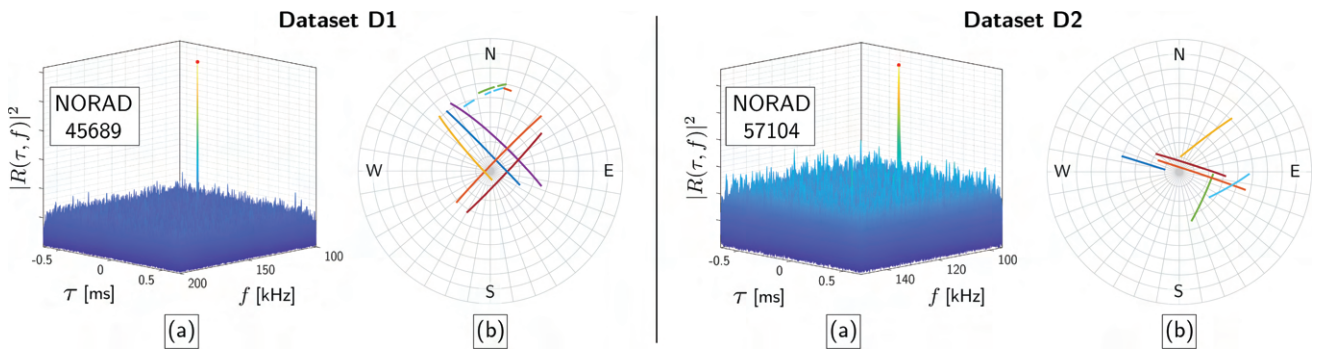


FIGURE 20 Data sets 1 and 2 – (a) Acquisition plots showing the detection of Starlink satellites using the OFDM beacon during the acquisition stage. (b) Skyplots showing the tracked Starlink satellites.

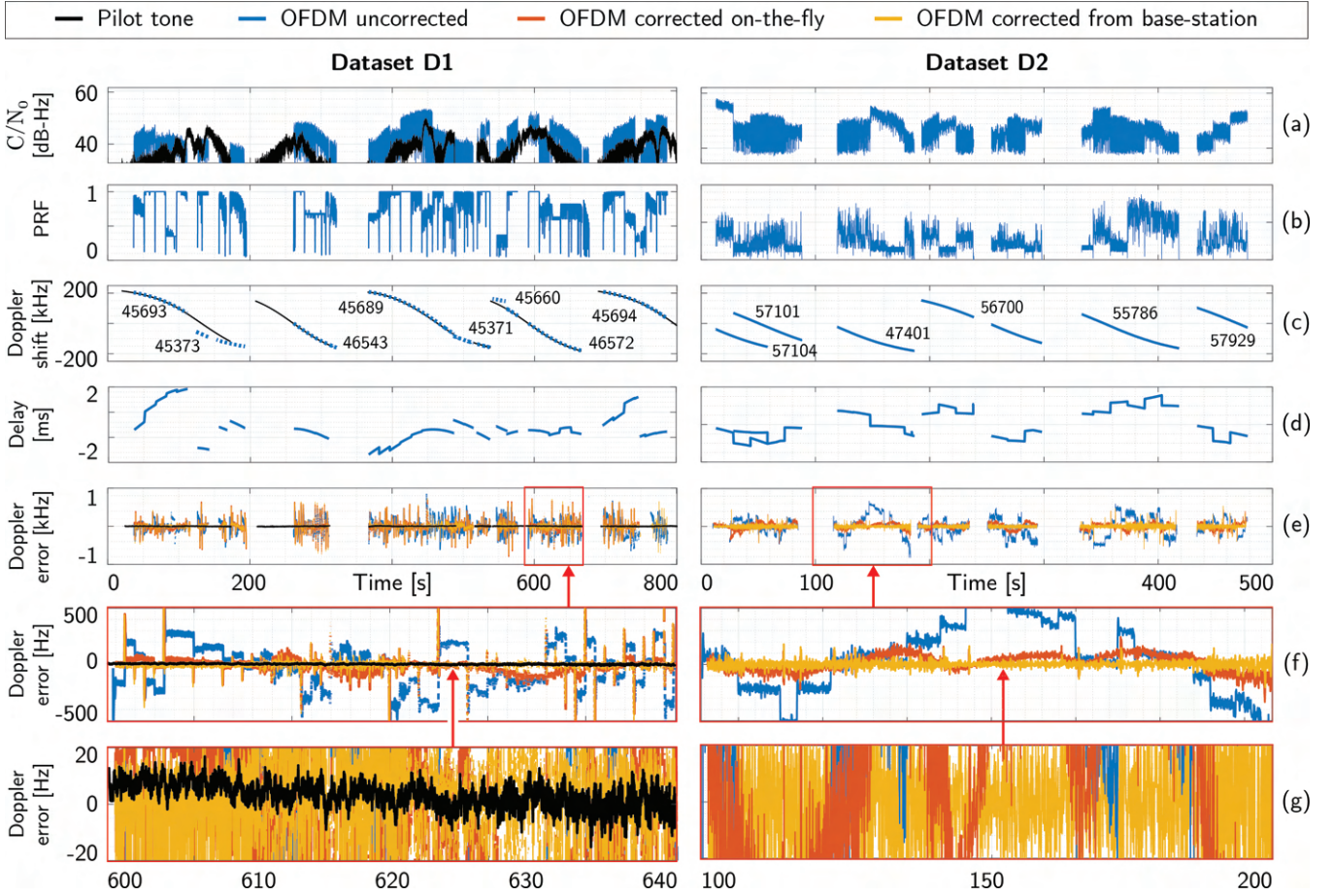


FIGURE 21 Data sets 1 and 2 – Starlink tracking results: (a) estimated C/N_0 curves of the tracked satellites based on the pilot tones beacon in black and OFDM beacon in blue; (b) PRF during OFDM transmission windows; (c) Doppler shift of the tracked satellites using the pilot tones and OFDM beacons; (d) code phase of the tracked satellites using the OFDM beacon; (e)–(g) error between the tracked Doppler and the Doppler calculated using the knowledge of the receiver's and satellite's positions. In (e)–(g), the pilot tones-based Doppler is shown in black, OFDM-based Doppler without corrections in blue, OFDM-based Doppler with on-the-fly correction in orange, and OFDM-based Doppler with correction communicated by a base station in yellow.

communicated by a base station in yellow. Finally, Figure 21(f) and Figure 21(g) show a close-up view of the error curves.

Overall, the tracking results demonstrate three main findings:

- While the method proposed in (Neinavaie & Kassas, 2024b) showed that only three of the six Starlink satellites were transmitting OFDM signals, Figure 21(c) shows that all six Starlink satellites (NORAD ID 45693, 46543, 45689, 45371, 46572, and 45694) are, in fact, transmitting OFDM signals.
- While previous literature showed the existence of only six satellites in data set D1 (Khalife et al., 2021; Kozhaya & Kassas, 2023; Neinavaie et al., 2021) by exploiting the pilot tones beacon, Figure 21(c) shows that two additional Starlink satellites are present in data set D1 but were transmitting only the OFDM beacon (NORAD ID 45373 and 45660).
- Some overhead satellites do not transmit pilot tones during OFDM transmission (e.g., Figure 21, NORAD ID 45373 and 45660).

8.2.2 | Positioning Results

Next, 200 MC realizations were used to initialize a WNLS estimator to localize the receiver. Figure 22 and Table 3 summarize the performance of the different positioning frameworks with Starlink satellites before year 2024. Given the assumed, TLE+SGP4-propagated satellites' position after correcting for temporal and orbital errors, the results showed that:

- The pilot tones-based Doppler shift (F1), which is not contaminated by the heavy CFO corrections, achieves positioning with meter-level accuracy.
- Using the OFDM-based Doppler shift in its raw form (F2) introduces noticeable estimation bias and variance, whereas estimating the CFO corrections on-the-fly (F3) improves the positioning performance to meter-level accuracy.
- Finally, correcting the OFDM-based Doppler using a cooperative base station that tracks the CFO correction history (F4) provides positioning results comparable to those achieved with pilot tones.

8.3 | Data Set 3

The third data set (D3) was collected in July 2024 on the roof of the ElectroScience Laboratory at The Ohio State University, Columbus, OH, USA. At the time of data collection, the pilot tones were not reliably detectable and trackable by LNBs, so

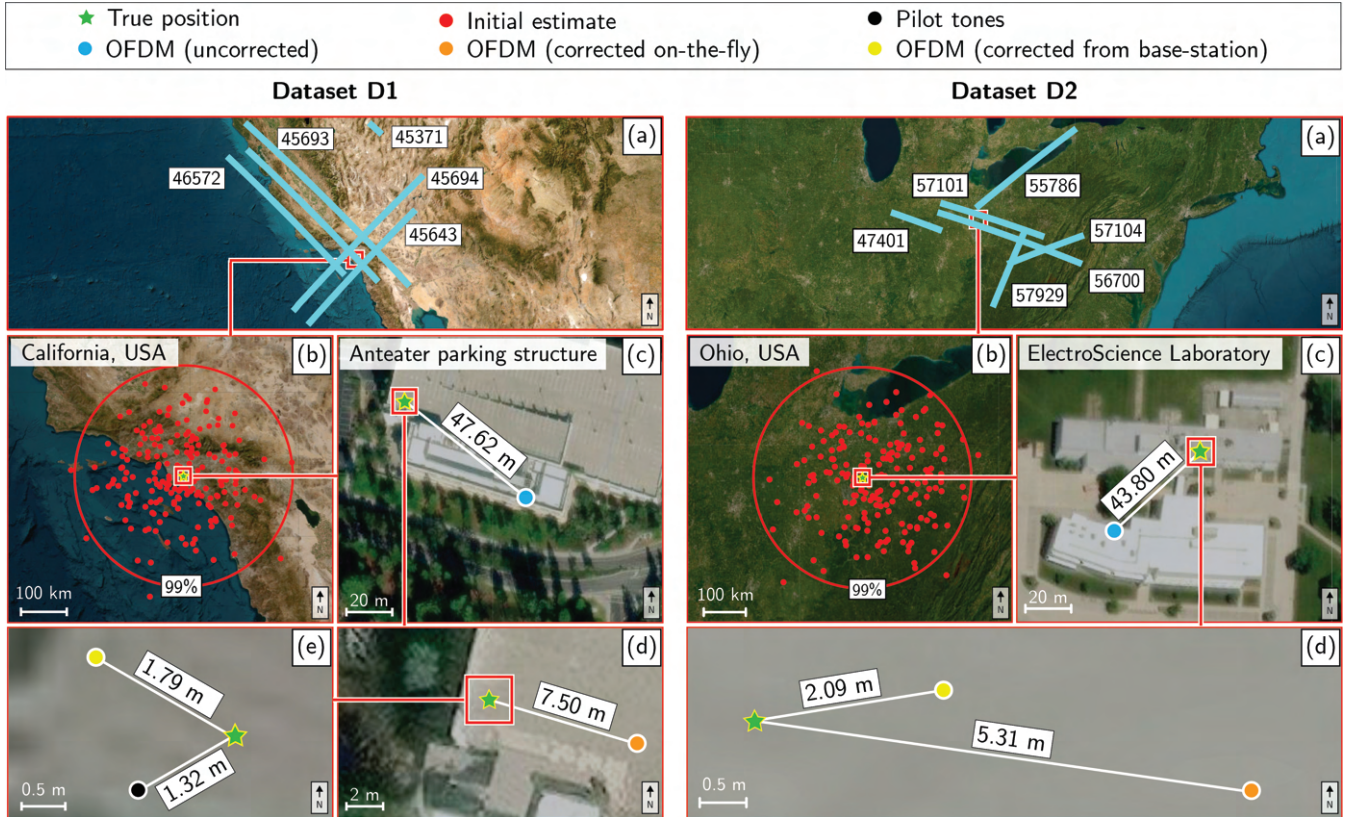


FIGURE 22 Data sets 1 and 2 – Starlink positioning results: (a) Starlink LEO satellite trajectories; (b) initial position estimate from 200 Monte Carlo realizations and the true position of the receiver; (c)–(e) average of the final estimated receiver's position using framework F1 in black, F2 in blue, F3 in orange, and F4 in yellow.

TABLE 3

Data Sets 1 and 2 – Comparison of Starlink Positioning Results: Average Error over 200 Monte Carlo Realizations.

Framework	Data Set D1		Data Set D2	
	2D error [m]	3D error [m]	2D error [m]	3D error [m]
Pilot tones	1.32	10.40	–	–
OFDM (uncorrected)	47.62	52.01	43.80	86.74
OFDM (corrected on-the-fly)	7.50	12.27	5.31	16.54
OFDM (corrected from assumed base-station)	1.79	6.97	2.09	13.8

only the OFDM beacon is exploited in this data set. Moreover, the OFDM-based Doppler in this data set does not exhibit Doppler biases. Unlike data sets 1 and 2, this data set was intended to maximize the exposure of the opportunistic receiver to the Starlink constellation by listening to all eight Ku-band downlink channels simultaneously. This approach drastically increased the number of visible satellites at any point in time, thereby enabling positioning capability as described below.

8.3.1 | Acquisition and Tracking

Over a 10-minute observation period, signals from 63 Starlink satellites were collected via an upward-pointing LNBF connected to a USRP 2955 and x 410 with a sampling rate of 2.5 MHz at the center frequencies of the eight Ku-band downlink channels. The two USRPs, each of four channels, were synchronized in both time and phase. The left plot of Figure 23 shows an acquisition of a Starlink satellite using the OFDM beacon during the acquisition stage. The right plot of Figure 23 shows a skyplot of all 63 tracked Starlink satellites.

After acquisition, the Starlink satellites were tracked using the OFDM beacon. Figure 24 shows the estimated C/N_0 , PRF, Doppler shift, code phase, and Doppler error curves of the tracked satellites. The PRF of the downlink shows moderate to high user activity in the serviced cell, which implies frequent measurement update rates. Note how, every 15 seconds, the satellite switches to a different channel frequency, the code phase exhibits a macro-correction, and the C/N_0 (power level) and PRF take different cluster values. When listening to all eight Ku-band downlink channels with an upward-facing LNBF, the average number of active Starlink satellites overhead was three.

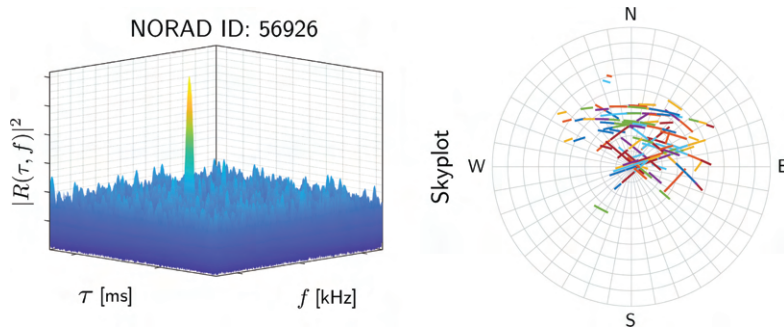


FIGURE 23 Data set 3 – *Left*: Acquisition plot showing the detection of a Starlink satellite using the OFDM beacon during the acquisition stage. *Right*: Skyplot showing the 63 tracked Starlink satellites.

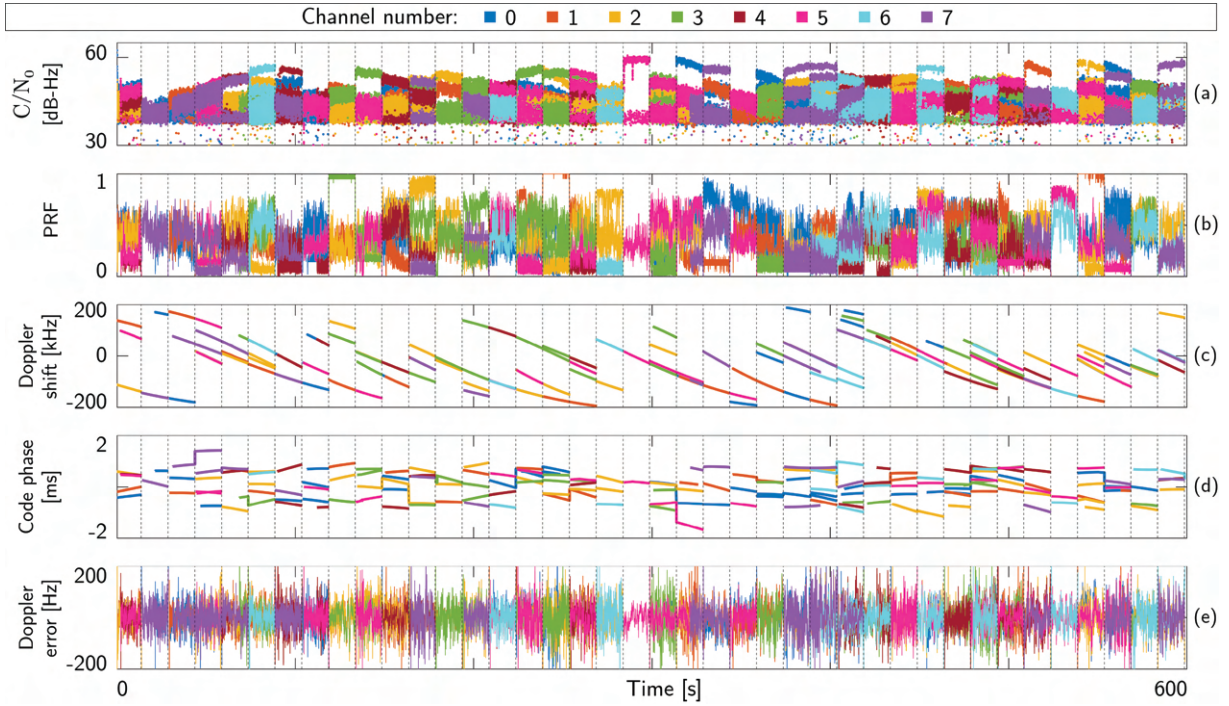


FIGURE 24 Data set 3 – Starlink tracking results: (a) C/N_0 curves of the tracked satellites; (b) PRF of the OFDM frames; (c) Doppler shift of the tracked satellites; (d) code phase of the tracked satellites; (e) error between the tracked Doppler and the Doppler calculated using the knowledge of the receiver's and satellite's positions.

8.3.2 | Positioning Results

Next, 200 MC realizations were used to initialize a WNLS estimator to localize the receiver. The batch window size was varied between 0.1 and 20 seconds and was swept throughout the duration of the data set with a step size of 0.1 seconds to generate a positioning solution using all of the satellites available for a given window. Figures 25 and 26 summarize the positioning performance.

Given the assumed TLE+SGP4 propagated satellites' position after correcting for temporal and orbital errors, and when an average of three Starlink satellites are available, a 90th percentile of two-meter error is achieved in 20 seconds, and a 90th percentile of 10-meter error is achieved in 8 seconds. These results indicate that the Starlink LEO satellites can support positioning in mobile applications with inertial aiding.

9 | CONCLUSION

This paper provided a comprehensive theoretical and experimental description for how to exploit Starlink LEO satellites for PNT.

First, the paper revealed, for the first time, the *full* Starlink OFDM beacon, which spans the whole time-frequency resource grid. This was achieved through *blind* beacon estimation, which showed that the Starlink sequences published in the literature only comprise 0.66% of Starlink's *full* OFDM. The paper additionally showed that exploiting the full OFDM beacon increases the receiver's process gain by nearly 18 dB compared to only using signals published in the literature.

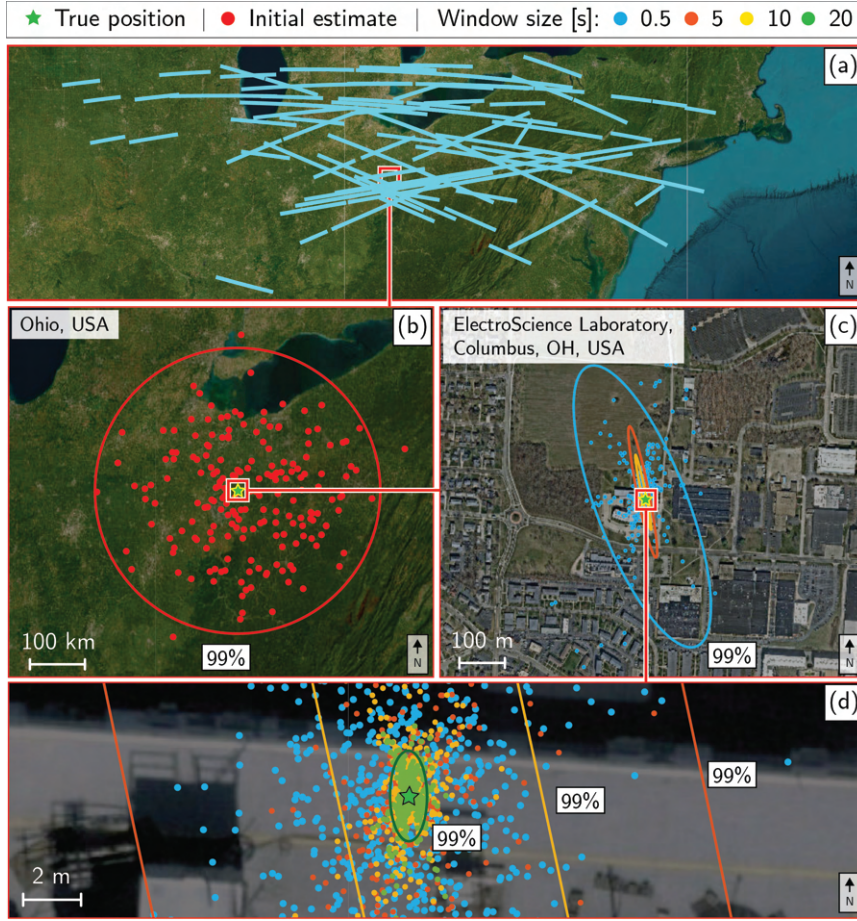


FIGURE 25 Data set 3 – Starlink positioning results: (a) Starlink LEO satellite trajectories; (b) initial position estimates from 200 MC realizations and the true position of the receiver; (c)–(d) final estimated receiver's 2D positions with the corresponding empirical 99th percentile ellipse for WNLS window sizes of 0.5, 5, 10, and 20 seconds in blue, orange, yellow, and green, respectively.

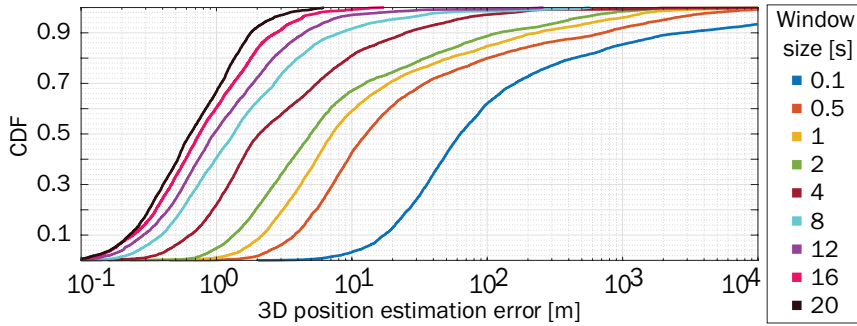


FIGURE 26 Data set 3 – Empirical cumulative distribution function of the 3D position estimation error of the WNLS as a function of window size.

Second, this paper studied and compared the maximum achievable received C/N_0 across different scenarios, namely: (i) pilot tones versus OFDM-based beacons and (ii) low-gain versus high-gain reception setups.

Third, this paper provided the first experimental results of extracting navigation observables using the OFDM signal (carrier phase, Doppler shift, and code phase) transmitted by Starlink satellites. The results showed that step-like corrections

contaminate the OFDM-based carrier and code phases from Starlink satellites, rendering their raw integration a challenge for precise and reliable PNT. However, these corrections are barely observed after year 2024, demonstrating how the structure and properties of proprietary signals can change over time.

Fourth, this paper showed how the consistent corrections made to the OFDM carrier frequency offset (CFO) can be estimated on-the-fly, with a good degree of fidelity, within the tracking loop of the software-defined receiver.

Finally, this paper demonstrated the first positioning solution obtained exclusively with OFDM-based Doppler shift. A 10-minute signal capture of the eight Ku-band downlink channels showed that an average of three active Starlink satellites are available at any time. Moreover, a positioning solution with meter-level 3D position root-mean-squared error can be achieved in 10 seconds from just three satellites.

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